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Twin contrastive interventional-cause hashing for unsupervised cross-modal retrieval

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ABSTRACT

Most unsupervised deep cross-modal hash retrieval (UDCMH) methods measure multimedia instances using similarity loss, while contrastive cross-modal hash retrieval (CMH) methods introduce contrastive loss. However, whether based on contrastive learning or not, these approaches are fundamentally similar, in which the contrastive learning module is plug-and-play. Meanwhile, they also have a problem of only using hashing as a semantic representation, lacking deeper analysis, and not integrating it into model construction or feature learning. In response to this dilemma, this paper presents twin contrastive interventional-cause hashing (TCICH) for unsupervised cross-modal retrieval. This represents the first attempt to incorporate both contrastive learning and hashing into model design, utilizing contrastive knowledge for group-wise causal reasoning. The binary opposite of the hash space can generate contrastive samples, and contrastive learning enhances the performance of hash generators. By employing the hash binary opposite value, we develop a twin contrast interventional-cause framework that uses contrastive dual sampling for data augmentation and proposes a new strategy for constructing an intrinsically interpretable UDCMH model. This model conducts interventions during training to create multiple interventional knowledge. Experiments on three baseline datasets demonstrate that the effectiveness of the proposed scheme surpasses that of most UDCMH methods. Given its comprehensive performance and innovation, our model design is well-suited for cross-modal retrieval tasks.

1. Introduction

The vast amount of multimedia data contains rich information, and cross-modal retrieval (CMR) can analyze correlations between different modal instances helping to alleviate the semantic gap caused by modality differences (Jie et al., 2025; Li & Li, 2024; Sun et al., 2024). In CMR research, several factors must be considered (Cui et al., 2024; Li et al., 2024; Wang et al., 2025). First, traditional mathematical methods struggle to analyze complex nonlinear correlations between different modalities (Kaur et al., 2021; Su et al., 2019; Tu et al., 2023), leading to the integration of deep learning technologies into most CMR methods. Second, for all retrieval application scenarios, timeliness and storage memory constrain the practicality of these models. Hashing technology can accelerate model training while maintaining performance at an acceptable level. Third, it is impractical to label vast amounts of cross-modal data, making unsupervised methods more viable and generalized than supervised methods. Based on these considerations, this paper pri-

marily explores unsupervised deep cross-modal hash retrieval (UDCMH) methods.

Currently, an essential bottleneck in most UDCMH methods (Xie et al., 2024; Yang et al., 2023) is effectively analyzing semantic information without given labels. For this issue, some cross-modal hash retrieval (CMH) methods focus on improving model design or training processes to effectively leverage cross-modal semantic similarities (Shi et al., 2022; Zhang et al., 2022). These improvements are typically analyzed from data characteristics, modal structure, sampling method, loss function, and other aspects (Duan et al., 2022; Zhao et al., 2021), but they may lack overall guidance for research direction. While all CMH methods employ hashing technology, to our knowledge, they only use hashing as a semantic representation, lacking deeper analysis, and not integrating it into model construction or feature learning. Therefore, our first starting point is to use the characteristics of hashing to assist model learning, providing macro and global guidance.

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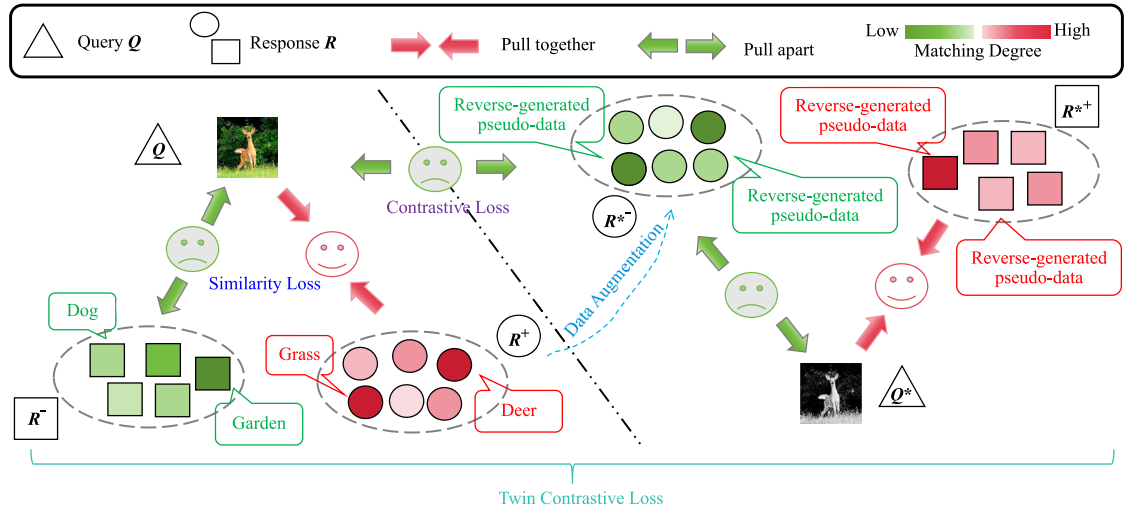


Fig. 1. The process of image-to-text retrieval (i.e., searching for texts through input images), which is analogous to text-to-image retrieval (i.e., searching for images through input texts). Most CMH paradigms use similarity loss to measure the relevance between a query and other samples, aiming to pull together related instances while separating unrelated ones. Through data augmentation, contrastive CMH methods introduce contrastive loss to assess pseudo-data against queries, thereby aiding model learning. However, from the perspective of sample learning and data augmentation, learning processes of similarity loss and contrast loss are essentially similar, that is, they all measure the differences between data (contrastive CMH methods add the contrastive learning module to implement semantic matching, but these designs not from the perspective of the overall model. In addition, these contrastive CMH methods only use hashing as a representation, not combining with contrastive learning. On their basis, by combining contrastive learning and hash technology, then using contrastive knowledge for group-wise causal reasoning, our proposed TCICH establishes the twin contrastive interventional-cause learning framework.

Meanwhile, to effectively explore semantic tags, some CMH methods introduced contrastive learning (Hu et al., 2022; Liang et al., 2024; Mikriukov et al., 2022; Wu et al., 2022a; Yang et al., 2022) in instances learning, such as Unsupervised Contrastive Cross-Modal Hashing (UCCH) (Hu et al., 2022). Most contrastive deep cross-modal hash retrieval (DCMH) methods apply contrastive learning to analyze the correlation with instances, making the distance between similar instances as small as possible and the distances between different instances as large as possible. However, the second issue arises here (or rather, the first issue: providing effective guidance for unsupervised models, has not been thoroughly resolved): the improvement of existing contrastive DCMH methods is gradual and lacks generalization. Specifically, these contrastive DCMH methods are essentially similar to the other non-contrastive DCMH methods, as shown in Fig. 1. In other words, for current CMH methods (whether based on comparative learning or not), the main focus is on analyzing the similarities and differences of samples (or guiding knowledge), limited to the discussion on data distribution, of which the innovations may be gradual, insufficient improvement on specific issues such as blurred boundaries between positive/negative samples, instability in cross-modal alignment. For this reason, our second starting point is to integrate contrastive learning into framework design and data augmentation, both model and instances are considered.

Therefore, unlike the other contrastive DCMH methods, we analyze contrastive learning from a new perspective, i.e., utilizing the binary opposition of hashing to perform data augmentation and constructing twin modules to achieve contrastive learning in the overall model. Specifically, (1) Hashing performs dimensional compression, converting sample features into the $[-1, +1]$ symmetric hash space. (2) In contrastive learning, it is necessary to use distance as the measure that the distances between similar features are as small as possible and vice versa. (3) Different from traditional methods, hash methods mainly use Hamming distance (and Cosine distance) to measure differences, instead of Euclidean distance and Markov distance. For example, for 4-bit hash codes $A = [+1, +1, -1, +1]$, the furthest point is $B = [-1, -1, +1, -1]$ by Hamming distance. Therefore, the samples represented by A and B are the most different in the whole hash space, the same goes for model learning. The binary opposite of the hash space can generate contrastive samples, and contrastive adversarial training can enhance the performance of hash

generators, making hash codes as distinct as possible within contrastive samples.

In this paper, we design twin contrastive interventional-cause hashing for unsupervised cross-modal retrieval (TCICH) and validate our proposed method on bidirectional image-text matching scenarios. First, we construct adversarial samples based on the binary antonymy of hash codes, i.e., data augmentation. In particular, our model generates Twin Hash Matrix W based on Hash Matrix B for subsequent training. Second, owing to the lack of transcendental guidance in unsupervised learning, many UDCMH methods currently establish Structural Joint Semantic Similarity Metric S , and we construct corresponding Twin Similarity Metric N to increase self-supervised information. Third, after reverse construction, we are attempting to construct interventional causality. TCICH uses interventional construction to utilize four submodules for causal reasoning. There is positive guidance in the inner subschemes, and reverse guidance in the inter subschemes. Finally, to our knowledge, we are the first to apply contrastive learning and interventional-cause learning with hashing in CMH model design, not limited to data augmentation. Therefore, our innovation can be extended to any hashing-based methods. The contribution of this paper can be summarized as follows:

- In response to the lack of effective guidance knowledge for UDCMH methods, a twin contrast framework that uses contrastive dual sampling for data augmentation and achieves group-wise contrastive learning to effectively utilize self-supervised information is proposed. Unlike other contrastive DCMH methods, we did not only apply contrastive learning in data feature labeling, but we fully studied the binary symmetry of hashing, of which the improvement is revolutionary and global.
- We propose a new strategy for constructing an intrinsically interpretable UDCMH model, which conducts interventions in the training process to create multiple interventional knowledge. Then, TCICH approaches the causal reasoning invariant across different knowledge while filtering out the unstable spurious response.
- In addition, we summarized the paradigm of self-supervised CMH methods, which can advance cross-media research including but not limited to causal learning. Adequate and comprehensive experiments conducted on three widely-used datasets verify that our proposed

TCICH can outperform most SOTA UDCMH algorithms and maintain good generalization.

In summary, we integrate contrastive learning with hashing from a new perspective, i.e., global design, and apply them to CMR tasks, achieving stable results in experiments. In addition, we use interventional construction to utilize four submodules for causal reasoning. By applying interventional-cause learning into the hash codes learning process, the model can keep training efficiency and enhance transferability across different scenarios, rather than being confined by environment-specific perturbation. We discussed TCICH from seven aspects on three datasets and used ablation experiments to analyze different components, while unsupervised frameworks ensured generalization. Although our current work focuses on CMR tasks, the integration of contrastive learning and hashing is for overall schemes, which has the opportunity to be extended to other hashing-based methods.

2. Related work

To effectively analyze multimedia data and complete CMR tasks, we briefly review some representative CMH works, including shallow (non-deep) (Ding et al., 2014; Kumar & Udupa, 2011; Song et al., 2013) and deep methods (Li et al., 2025; Su et al., 2019; Zhu et al., 2022). Deep methods can be based on supervised learning or unsupervised learning. In addition, we introduce the application of contrastive learning in CMH methods (Cui et al., 2024; Hu et al., 2022).

2.1. Shallow (Non-deep) cross-modal hash retrieval methods

As the first batch of research to study CMR tasks, shallow CMH methods mainly analyze based on the distribution of cross-modal instances and mathematical statistical knowledge. Cross-View Hashing (CVH) (Kumar & Udupa, 2011) attempted to analyze the similarity of samples from different views and then project it into a public space. Inter-Media Hashing (IMH) (Song et al., 2013) used the low-latitude regression scheme to analyze the correlation of samples from different modalities. Collective Matrix Factorization Hashing (CMFH) (Ding et al., 2014) proposed multi-modal collective matrices and factorization frameworks to complete CMR tasks. Latent Semantic Sparse Hashing (LSSH) (Zhou et al., 2014) designed sparse hashing and latent semantic models to analyze cross-modal data structures. The advantage of these shallow methods is that they have complete logical reasoning and good time complexity, but their ability is unsatisfactory to handle modality nonlinearity correlations (Kaur et al., 2021; Su et al., 2019; Tu et al., 2023; Wang et al., 2025). Meanwhile, considering the size and complexity of current datasets, relying solely on traditional mathematical technologies may be outdated and inefficient. Therefore, to better characterize nonlinear features and process large-scale data, the deep learning-based method is more widely used.

2.2. Deep cross-modal hash retrieval methods

Deep fine-grained classifiers can easily mine the knowledge contained in samples from images, videos, etc., and can efficiently process large-scale data. Therefore, DCMH methods are currently the mainstream direction (Huang et al., 2024; Li et al., 2023; Yao et al., 2024). Unsupervised Generative Adversarial Cross-Modal Hashing (UGACH) (Zhang et al., 2018) proposed a generative adversarial network for modeling cross-modal hashing in an unsupervised fashion. Cycle-Consistent Deep Generative Hashing (CYC-DGH) (Wu et al., 2018a) can produce hash embeddings in the absence of paired training correspondence and realize cross-modal semantic alignment. Deep Joint Semantics Reconstructing Hashing (DJSRH) (Su et al., 2019) analyzed the distribution of multi-modal data and designed the auxiliary similarity matrices to help model training. Joint-modal Distribution-based Similarity Hashing (JDSH) (Liu et al., 2020) conducted a more in-depth analysis of the characteristics of the samples and designed the joint-modal

model. Deep Graph-Neighbor Coherence Preserving Network (DGCPN) (Yu et al., 2021) analyzed the sample features that need to be learned, and used graph models to represent the relations between samples to complete model training. Correlation-Identity Reconstruction Hashing (CIRH) (Zhu et al., 2022) constructed a multi-modal collaboration graph to model intra-modal and inter-modal correlations, which can improve matching performance. High-Order Nonlocal Hashing (HNH) (Zhang et al., 2021) proposed a high-order affine measure that represents cross-domain neighborhood structures from a non-local perspective, to characterize cross-modal features comprehensively. Label-Consistent Kernel Transform Learning based Sparse Hashing (LcKTLSh) (Maggu et al., 2025) added semantic mapping transformation in sample projection and introduced label consistency in the overall model learning, which effectively improves the quality of knowledge alignment. It is worth noting that some cross-modal retrieval methods in other scenarios are also representative, such as Cross-modal Contrastive Learning for Multi-modal Video Representations (CrossCLR) (Zolfaghari et al., 2021), which proposed a contrastive loss and filtered their input embeddings to avoid false negative issues. These deep methods have thoroughly analyzed the features of multi-modal data and typically have good experimental performance.

2.3. Contrastive learning

Supervised DCMH methods (Li & Li, 2024; Zhu et al., 2022) are limited in practical applications because of the inability to label all samples in real life; UDCMH methods (Liu et al., 2020; Su et al., 2019) may lack effective guidance knowledge in model learning. So some DCMH methods applied contrastive learning to optimize model design and improve matching performance. We also introduced some representative contrastive-based UDCMH methods later. Contrastive learning (Cai et al., 2020; Chen et al., 2020; Hadsell et al., 2006; He et al., 2020; Khosla et al., 2020; Tian et al., 2020; Wu et al., 2018b) is a very effective unsupervised learning technique, models can effectively tap into knowledge and be trained through self-supervised learning. Contrastive Label Correlation Enhanced Unified Hashing Encoder (UniHash) (Wu et al., 2022a) proposed a shared transformer encoder that incorporates comparative learning. Deep Unsupervised Cross-modal Contrastive Hashing (DUCH) (Mikriukov et al., 2022) compared targets that can maintain similarity within different modalities in remote sensing. Cross-Modal Mutual Quantization (CMMQ) (Yang et al., 2022) designed a proxy-based contrastive loss to reduce the impact of noise labels and heterogeneous gaps. Unsupervised Contrastive Cross-Modal Hashing (UCCH) (Hu et al., 2022) proposed the cross-modal ranking learning loss approach that used contrastive learning for positive and negative samples to improve model performance. Structure-Aware Contrastive Hashing (SACH) (Cui et al., 2024) used high-dimensional and discriminative network learning representations to represent richer semantic information. To better explore the potential structural information and achieve high-quality semantic alignment, Graph Contrastive-and-Reconstructive Hashing (GCRH) (Wei et al., 2025) designed global intra-modal and inter-modal graph contrastive learning, and then proposed local graph reconstruction to be integrated into the model design together.

In addition, with the development of contrastive learning, multi-level contrastive mechanisms have also been studied and applied in many fields, such as knowledge tracing and multimodal sentiment analysis, etc. Multi-Level Graph Contrastive Learning (MLGCL) (Shao & Tao, 2024) framework learned robust representations of graph data by comparing spatial views, using a multi-level contrastive pattern to simultaneously maintain local and semantic similarity. Multi-Level Contrastive Learning and Multi-Layer Convolution Fusion (MCL-MCF) method (Fan et al., 2024) proposed three levels of multimodal sentiment analysis: the first and second levels primarily focus on addressing the heterogeneity of basic information features, while the third level primarily addresses high-level semantic information features. Multi-level Contrastive Learning model for Knowledge Tracing (MCKT) (Shen et al., 2025) established

Table 1

A brief comparison of contrastive DCMH methods. S represents supervised methods, U represents unsupervised methods. Loss function can be divided into using InfoNCE (van den et al., 2018) (or partial) and other native proposed designs.

Method	S/U	Loss (or partial)	Essential Design
UniHash (Wu et al., 2022a)	S	InfoNCE	Using contrastive learning to achieve negative label generation.
DUCH (Mikriukov et al., 2022)	U	InfoNCE	Applicable to remote sensing scenarios, introducing cross entropy objective.
CMMQ (Yang et al., 2022)	S	Other	Construct a proxy-based contrastive loss function.
UCCH (Hu et al., 2022)	U	InfoNCE	Build the contrastive dictionary sets for ranking.
SACH (Cui et al., 2024)	U	InfoNCE	Adopting decoupled contrastive learning (DCL) (Yeh et al., 2022) to align high-order semantic information.
GCRH (Wei et al., 2025)	U	Other	Integrate the latent graph structural information provided into contrastive learning.
TCICH	U	Other	Combining hash representation with contrastive learning.

multi-level contrastive learning from the problem level, interaction level, and knowledge state level to enhance the advantageous representation of knowledge tracing. Multi-Level Contrastive Learning (MLCL) for multimodal sentiment analysis framework (Zhuang et al., 2025) consists of three parts: uni-modal, bi-modal, and tri-modal contrastive learning, which aligns the non-overlapping uni-modal and bi-modal representations and emphasizes the complementarity of tri-modal information. There are also some methods that attempt to apply multi-level contrastive mechanisms to hybrid image retrieval, Multi-Level Contrastive Learning (MLCL) method (Zhao et al., 2024) combined mixed query features into fused features through cross-pattern contrastive learning and multi-layer semantic alignment. Although these explorations of multi-level contrastive learning can provide novel research ideas, they are currently not applicable to cross-modal hash retrieval scenarios, so only appropriate comparative discussions were conducted.

For specific CMR tasks, to better compare the differences between existing contrastive DCMH methods, we organized Table 1 to analyze them from three aspects: label correction, loss construction, and essential design. Existing methods have conducted several studies on integrating contrastive learning into cross-modal schemes from multiple perspectives. However, many work only transfer contrastive learning to cross-modal scenarios, lacking specialized design. This differs significantly from our strategy, as no work combines the characteristics of hash space structure with contrastive learning.

2.4. Existing limitations

Based on previous discussions and Table 1, despite attempting multiple strategies and achieving preliminary cross-modal matching, the existing methods still have significant limitations. Firstly, these contrastive DCMH methods are essentially similar to other non-contrastive DCMH methods. Specifically, most current contrastive DCMH methods (Hu et al., 2022) integrated contrastive learning into cross-modal hashing, fully considered data distribution, and performed discrete optimization on the similarity matrix to bridge the heterogeneity gap, these improvements seem also to be achievable without utilizing contrastive learning.

Secondly, beyond the intrinsic research of contrastive learning in cross-modal hashing learning, there are also some specific, unresolved challenges in existing hashing-based retrieval methods. The semantic representation compression introduced by the hash binary space may result in blurred boundaries between positive and negative samples, and the overall sample representation may be limited to a narrow range. Additionally, there is the semantic loss that occurs during the hash compression process, which can lead to instability in cross-modal alignment. The reason these issues exceed the current work's ability (or cannot be completely solved) is that hash representation is only used as a semantic compression in their scheme, without exploring the underlying mechanisms and designing models based on the hashing structure.

In addition, multi-level contrastive mechanisms hold great potential in cross-modal matching, but their practical implementation requires further in-depth research. In summary, the key challenge in current research is that the existing contrastive CMH work is limited to superficial environment-specific data augmentation and lacks consolidation of other factors, such as hash representation.

On the basis of the shortcomings of the current work, we fully analyzed the relationship between hash technology and contrastive learning, then used contrastive knowledge for group-wise causal reasoning, i.e., twin contrastive interventional-cause learning. Specifically, in response to the attenuation and instability of semantic representation, we designed contrastive double sampling to enhance the number of samples for implementing sample augmentation. Then, causal intervention was introduced in the model design, attempting to enhance the robustness and reduce the interference of unstable factors. Besides, we also drew on some recent foundational contributions (Maggu & Majumdar, 2023; Maggu et al., 2020b) in transform learning and transformed clustering, which provide some ideas in creating structured and semantically separable latent spaces. Although some methods (Wang et al., 2019; Yang et al., 2024) have similar starting points to ours, we designed unique and more innovative implementation solutions. To our knowledge, we are the first work to apply contrastive learning and interventional-cause learning with the hash principle in CMH model design, not limited to data augmentation.

3. Methodology

3.1. Contrast of hashing

Hashing converts numerical values into binary ranges, such as $[-1, +1]$ or $[0, +1]$. As shown in Fig. 2(a), the value of a 1-bit hash code can be transformed into a straight line: A (-1) and B (+1) are the least correlated. The situation expanded to 2-bit is shown in Fig. 2(b), which can represent more situations. Compared with B, A is more related to D (or C). When we continue to increase the hash length, excessive elements will reduce the readability of the image, so we omitted other points and the coordinate axis in Fig. 2(c) to form Fig. 2(d). At this time, we noticed that the symmetry of the hash space could be used to construct contrastive learning. For example, for Fig. 2(d), it can be considered that when the hash code of one sample is A, we reverse generate B for contrastive learning. The train-generated hash matrix is strictly binary, however, many CMH methods establish spatial semantic similarity for instances in the training process (Liu et al., 2020; Su et al., 2019). For the sake of convenient calculation, the spatial semantic similarity space is the same as the hash space. When the spatial semantic similarities of some samples are projected to the hash space, which can be shown in Fig. 2(e). These points will not be completely binary but still between values $[-1, +1]$, so these guide knowledge can be reused in reverse. Through these analysis, we can integrate the contrast of hashing into model design and data augmentation. For data augmentation, specifically, we use the Hamming distance to reverse construct, as shown in Fig. 2(f), using the original samples as the positive instances, and the reverse samples as the negative instances. From some representative theories (Carreira-Perpinán & Raziperchikolaei, 2015; Kulis & Darrell, 2009), it is feasible that binary-opposite hash codes are guaranteed to act as informative hard negatives. In the following sections, we will further discuss the issues that CMH methods need to handle, and the integration of contrastive learning with hashing in model design.

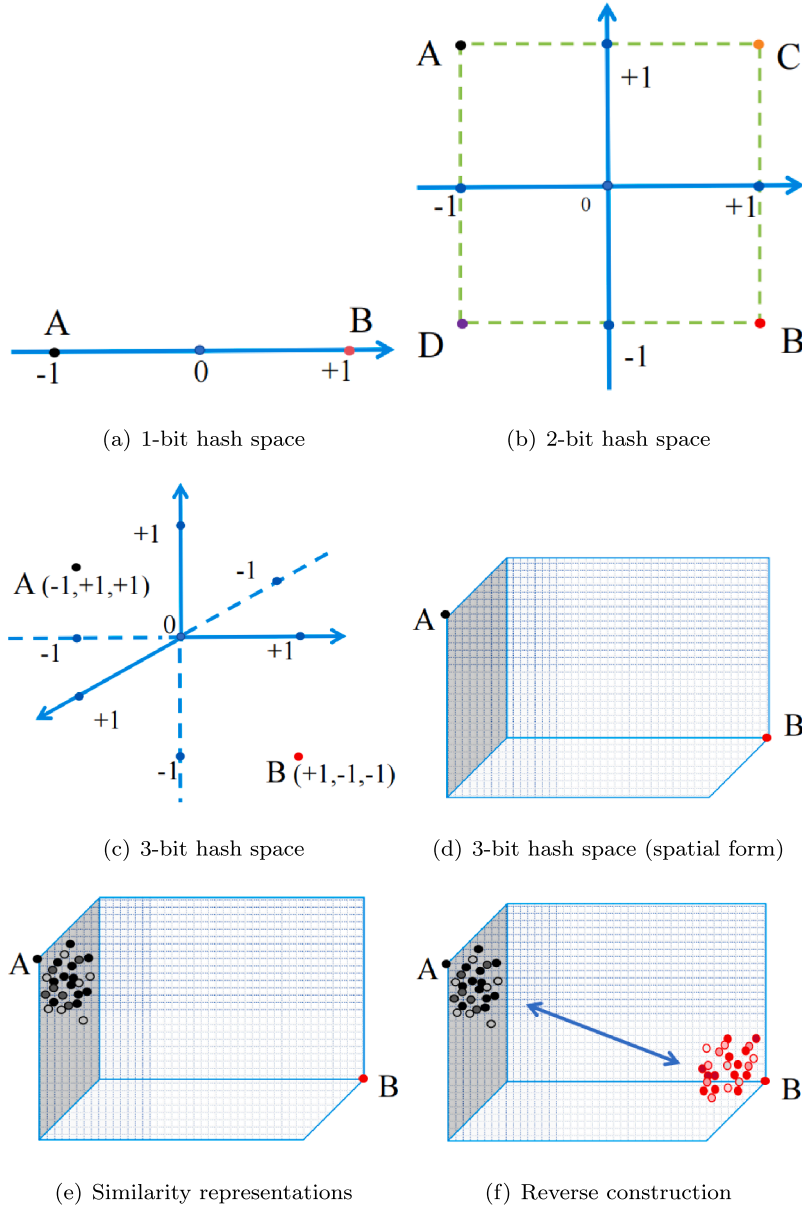


Fig. 2. Different from real space, hash space is symmetrical and discrete, usually measured using the Hamming distance. Fig. 2(a), (b), and (c) respectively show the number of classification points that can be represented by hash spaces in different lengths. After converting the three-dimensional coordinate form into spatial form, as shown in Fig. 2(d), point A and point B are the opposite points with the maximum Hamming distance (most dissimilar). For some data points within a certain hash classification space, we can use the hash binary opposition to construct negative data points for contrastive learning, as shown in Fig. 2(f).

3.2. Problem definition

Before discussing our twin contrastive hashing framework, we first introduce the issues studied. In real life, there is so much multimedia information from various modalities, these data come from different modalities, but they may have similar meanings. How to effectively utilize the correlation between cross-modal data and improve retrieval accuracy, which is the goal of CMR research. Without loss of generality, we focus on retrieval tasks on bimodal data (i.e., image and text) in this paper. In the image-text dataset $\mathbf{D} = \{\mathbf{d}_x\}_{x=1}^n = \{\mathbf{i}_x, \mathbf{t}_x\}_{x=1}^n$, $\mathbf{i}_x$ and $\mathbf{t}_x$ denote the original image-text pairs for the $\mathbf{d}_x$. Typically, $\mathbf{d}_x$ is not always related to $\mathbf{d}_y$, so similarity matrix $\mathbf{C}$ (in which $C_{xy} \in \{0, 1\}$) be introduced in CMR tasks to characterize the correlation. In similarity matrix $\mathbf{C}$, if $C_{xy} = 1$, which means that $\mathbf{d}_x$ is related to $\mathbf{d}_y$, otherwise $C_{xy} = 0$. Differ from CMR methods, CMH methods extra need to transform these features into unified binary codes $\mathbf{B} = \{\mathbf{b}_x\}_{x=1}^n \in \{-1, +1\}^{n \times k}$, k is

the length of hash codes. In each batch training, our TCICH randomly samples subset $\phi \in \mathbf{D}$ and set $\mathbf{B} = \{\mathbf{b}_x\}_{x=1}^m \in \{-1, +1\}^{m \times k}$, m is the size of the random sampling subset. The generated hash codes will be used to represent the correlation of the samples, without considering the modalities gap. The efficient CMH model will generate similar hash codes for samples with similar semantics (Hamming distance is small), otherwise, the Hamming distance is expected to be as large as possible.

3.3. Preprocess modules

As illustrated in Fig. 3, our TCICH has four modules, namely, cross-modal hashing learning, structural semantic similarity excavation, reverse construction, and twin contrastive learning. This section mainly introduces two preprocess modules, i.e., cross-modal hashing learning and structural semantic similarity excavation. To demonstrate Fig. 3 more clearly, we use green to represent text instances, purple to

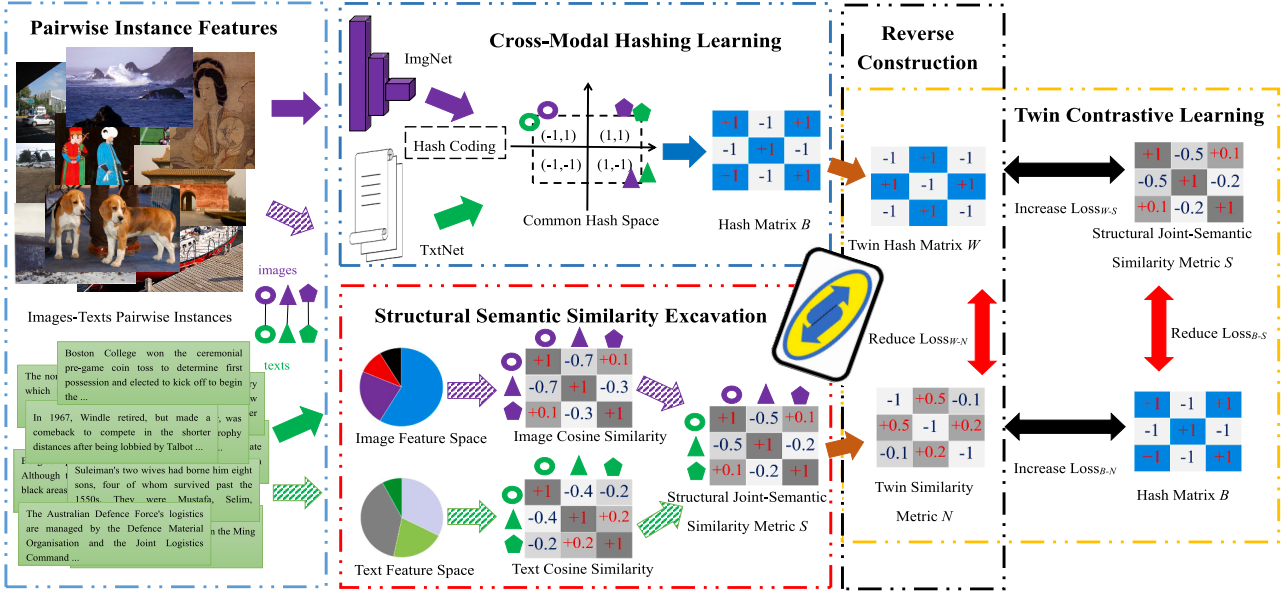


Fig. 3. The framework of our TCICH. In the preprocess modules, ImgNet and TxtNet extract the basic features of image-text pairwise instances to generate Hash Matrix **B** and Structural Joint Semantic Similarity Metric **S**. Then, we combine the symmetry of hash space and Hamming distance to reverse construct Twin Hash Matrix **W** and Twin Similarity Metric **N**. Finally, we design twin comparative interventional-cause learning using contrastive knowledge for group-wise causal reasoning. Specifically, there is positive guidance in the inner subschemes, while reverse guidance is in the inter subschemes, the four loss functions have different effects in model training.

represent image instances, and the same semantic pair represents the same shape. In the semantic preprocess modules, solid arrows represent the process of cross-modal hashing learning, and dashed arrows represent the process of structural semantic similarity excavation. In the reverse construction stage, "UNO card" and orange arrows indicate reverse generation. At the right, i.e., twin contrastive learning, the red arrows represent pulling closer, and the black arrows represent pushing farther.

For original pairwise instances, the first part of all CMH methods needs to convert them into hash space for subsequent training. The backbone of ImgNet is usually pre-trained CNNs or Transformers (Vaswani et al., 2017), and the backbone of TxtNet is usually Bag-of-words (BOW) or Bert (Devlin et al., 2018). Multimedia instances are transformed through ImgNet and TxtNet to project in the common hash space, and then generate Hash Matrix **B**. This paper sets the output of the last hidden layer as $\mathbf{F}$, specifically, $\mathbf{F}_I \in \mathbb{R}^{m \times p_I}$ is p_I -dimensional representation extracted by ImgNet, and $\mathbf{F}_T \in \mathbb{R}^{m \times p_T}$ is p_T -dimensional representation extracted by TxtNet. The binary hash codes will be generated by

$$\mathbf{B} = \text{sign}(\mathbf{F}) \in \{-1, +1\}^{m \times k}, \quad (1)$$

where $\text{sign}(\cdot)$ function is used to convert the input value, the positive numbers are +1, and the others are -1. In the process of BP, for all non-zero inputs, the gradient of the $\text{sign}(\cdot)$ function will be zero, which will destroy the gradient and make it impossible for the gradient to return to the top. To solve the problem of gradient vanishing (Li & Li, 2024; Su et al., 2019), this paper uses the $\tanh(\cdot)$ function to generate hash codes and conducts approximate convergence, the reason is that the $\tanh(\cdot)$ function can be used to approximate the $\text{sign}(\cdot)$ function, i.e.,

$$\lim_{u \rightarrow +\infty} \tanh(ux) = \text{sign}(x) \quad (2)$$

and is more suitable for deep learning. So we use

$$\mathbf{B} = \tanh(\mathbf{F}) \in \{-1, +1\}^{m \times k} \quad (3)$$

in the actual training process. $\mathbf{B}_I$ and $\mathbf{B}_T$ are hash codes generated by image $\mathbf{x}_i$ and text $\mathbf{t}_j$ through the ImgNet and TxtNet, respectively. $\mathbf{B}_I$ and $\mathbf{B}_T \in \{-1, +1\}^{m \times k}$, k represents the hash code length.

Another preprocess module is structural semantic similarity excavation, of which the purpose is to construct effective guiding knowledge through self-supervised learning. Our work focuses on contrastive learning, hash representation, and causal intervention reasoning, then presents a novel design for cross-modal matching tasks. To avoid overly cumbersome design and highlight our essential contribution, we adopt a classic strategy for measuring semantic similarity. To generate image semantic similarity metric $\mathbf{S}_I$, we normalize $\mathbf{F}_I$ to $\hat{\mathbf{F}}_I$, and $\mathbf{S}_I = \hat{\mathbf{F}}_I \hat{\mathbf{F}}_I^T \in \{-1, +1\}^{m \times m}$. Similarly, $\mathbf{F}_T$ can be normalized to $\hat{\mathbf{F}}_T$, and text semantic similarity metric $\mathbf{S}_T = \hat{\mathbf{F}}_T \hat{\mathbf{F}}_T^T \in \{-1, +1\}^{m \times m}$. Therefore, Structural Joint Semantic Similarity Metric **S** can be defined as follows:

$$\mathbf{S} = (1 - \delta)[\theta \mathbf{S}_I + (1 - \theta) \mathbf{S}_T] + \frac{\delta}{n} [\theta^2 \mathbf{S}_I \mathbf{S}_I^T + \theta(1 - \theta) \mathbf{S}_I \mathbf{S}_T^T + \theta(1 - \theta) \mathbf{S}_T \mathbf{S}_I^T + (1 - \theta)^2 \mathbf{S}_T \mathbf{S}_T^T], \quad (4)$$

where δ and θ are the trade-off parameters for adjusting the weight of the high-order neighborhood measure. Many UDCMH methods train **S** as self-supervised knowledge for guide semantic-related hash codes learning (Liu et al., 2020; Su et al., 2019). The heatmaps of the instances after structural semantic similarity excavation are shown in Fig. 4.

From the heatmaps visualization results, although this feature processing performance is difficult to compare with some advanced schemes (Li et al., 2019, 2022), the similarity extraction design is relatively effective and can obtain preliminary matching knowledge. Through previous processing, the training objective of these methods is as follows:

$$\text{loss} = \min_{\mathbf{B}_I, \mathbf{B}_T} \|\gamma \mathbf{S} - \cos(\mathbf{B}_I, \mathbf{B}_T)\|_F^2 + \mu_1 \|\gamma \mathbf{S} - \cos(\mathbf{B}_I, \mathbf{B}_I)\|_F^2 + \mu_2 \|\gamma \mathbf{S} - \cos(\mathbf{B}_T, \mathbf{B}_T)\|_F^2, \quad (5)$$

where γ is a hyper-parameter and is used to adjust the weight of **S**, μ_1 and μ_2 are the trade-off parameters for balancing the inter-modal and intra-modal reconstructions, the specific values of μ_1 and μ_2 can be regulated in the experiments. To more effectively implement the learning performance, we constructed the twin contrastive hash schemes. Now, it is time to reverse.

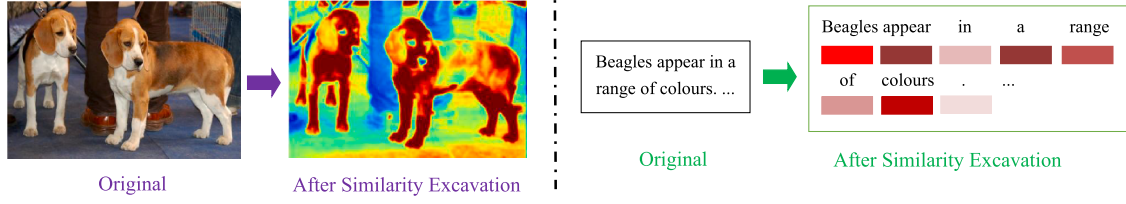


Fig. 4. The heatmaps of the instances after structural semantic similarity excavation. In image heatmap, the region of interest with the generated bounding boxes covers almost the entire image in a relatively dispersed manner. For text heatmap, darker colors represent higher attention weights.

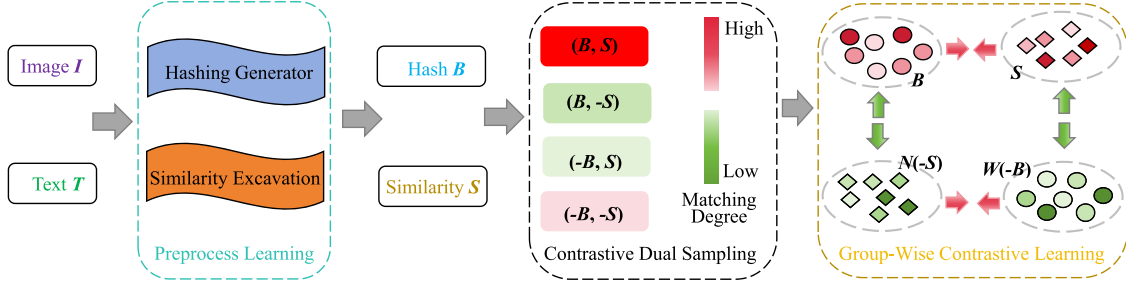


Fig. 5. Through preprocess learning, we convert the input Image I and Text T into Hash Matrix B and Structural Joint Semantic Similarity Metric S . Then after contrastive dual sampling, Twin Hash Matrix $W(-B)$ and Twin Similarity Metric $N(-S)$ are generated. In addition to data augmentation like other contrastive DCMH methods, we are attempting to construct interventional causality.

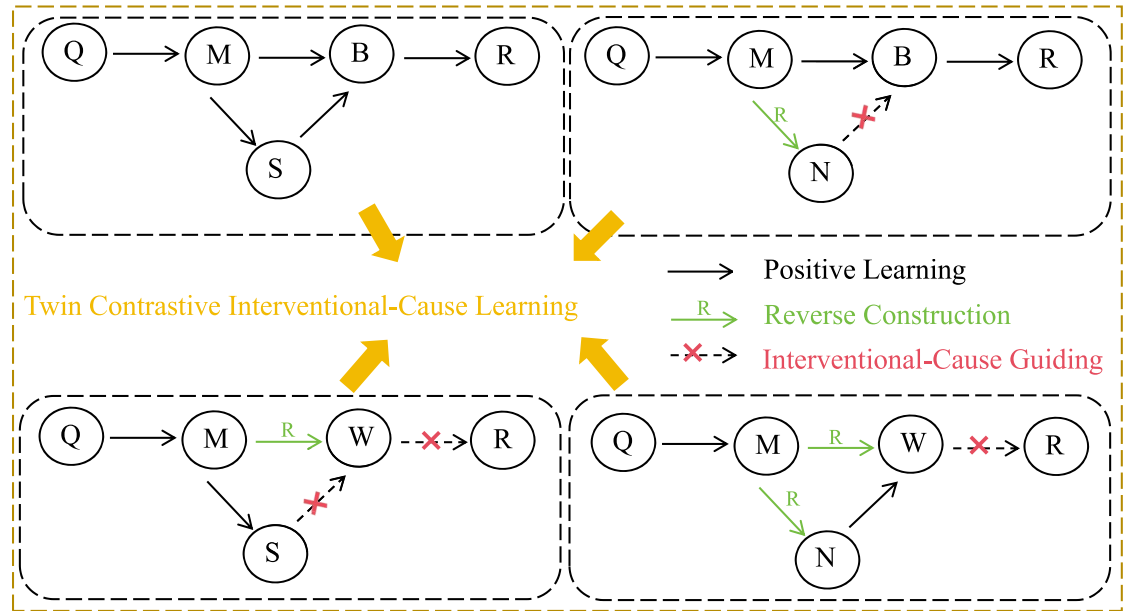


Fig. 6. Causal view of twin contrastive interventional-cause learning.

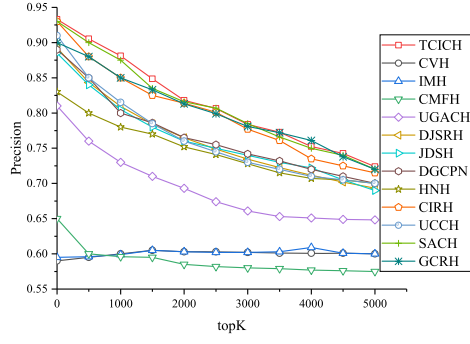
3.4. Reverse construction (Contrastive dual sampling)

As shown in Fig. 5, we implement data augmentation in the hash space and construct Twin Hash Matrix W through B first, i.e., $W = -B$. This approach has two benefits, one is data augmentation. In addition, hash codes represent the sample classification by pre-trained networks. By reverse construction, one sample will have two classification hash codes. In subsequent training, the positive classification will reward the model and the negative classification will punish the model. Therefore, the other benefit is the BP of W can enhance the performance of the hash-generated networks.

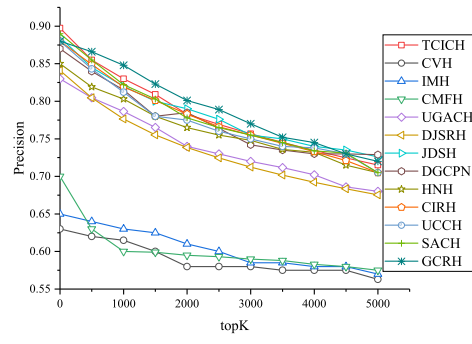
The purpose of establishing S is to perform positive proofreading on B , to effectively utilize S , we built the Twin Similarity Metric N . As mentioned earlier, S is not strictly binary and N can be constructed in

various ways, such as negation and remainder. Different construction methods correspond to different model loss function designs. We are planning to use N in the twin scheme to provide positive guidance to W , so we set $N = -S$. After reverse construction, our TCICH model has four essential hash components, as shown on the right side of Fig. 3, divided into two frameworks: B and S are the original frameworks, and W and N are the twin frameworks. This self-interference occurs in the early training stages, but is beneficial for robustness (Korb et al., 2004; Li & Li, 2026).

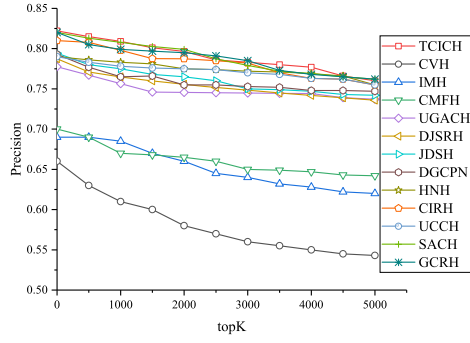
In summary, as shown in Fig. 5, reverse-generated pseudo-metrics and native matrices shave group-wise contrastive learning, including four sub-relationships, which effectively combine contrastive learning and hash in model construction. By analyzing some latent representations designed for retrieval tasks and the attempt at interventional



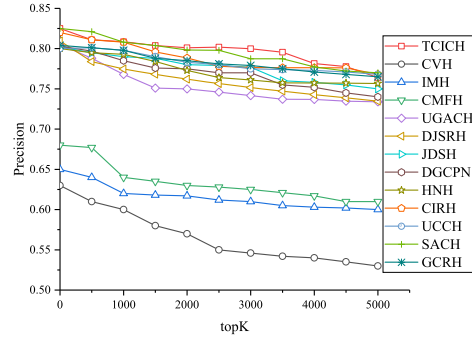
(a) I2T@MIRFLICR-25K(128-bits)



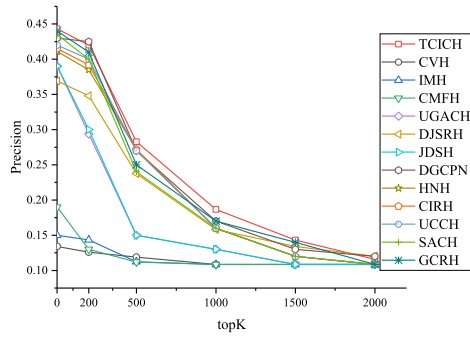
(b) T2I@MIRFLICR-25K(128-bits)



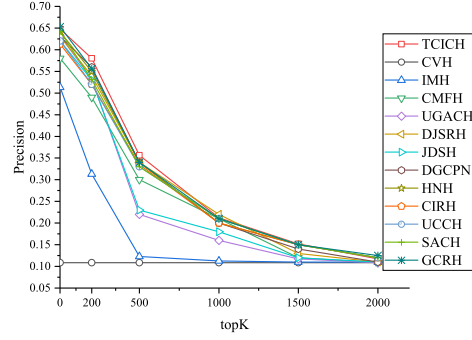
(c) I2T@NUS-WIDE(128-bits)



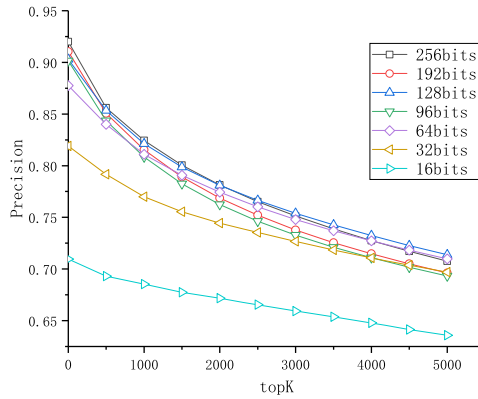
(d) T2I@NUS-WIDE(128-bits)



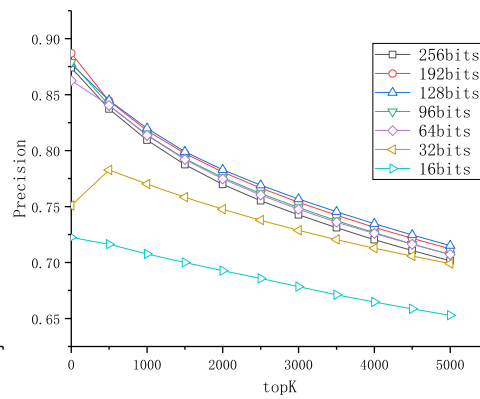
(e) I2T@Wikipedia(64-bits)



(f) T2I@Wikipedia(64-bits)

Fig. 7. The $P@topK$ curves of different methods.

(a) I2T@MIRFLICR-25K



(b) T2I@MIRFLICR-25K

Fig. 8. The $P@topK$ curves at different hash codes length.

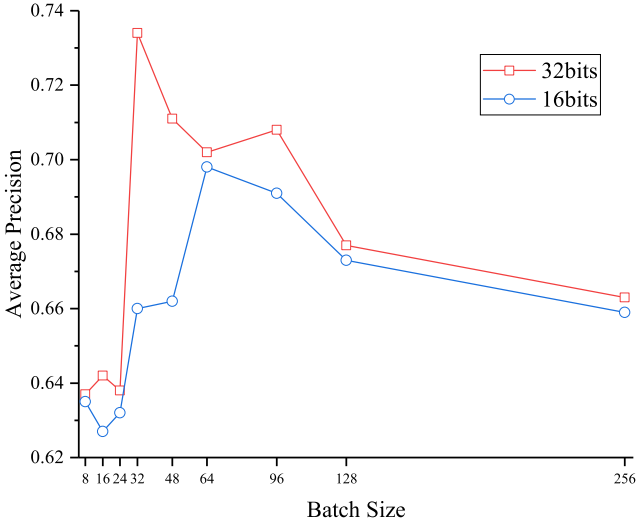


Fig. 9. Average precisions of different batch sizes on MIRFlickr-25K datasets when short hash lengths.

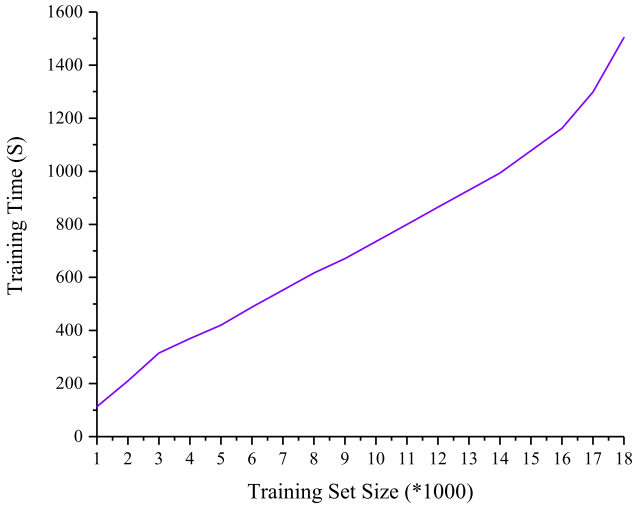


Fig. 10. Training time on MIRFlickr-25K dataset with different training set size.

learning objectives (Maggu & Goel, 2024; Maggu et al., 2020a), TCICH provides one sample for two classifications of hash codes from the hash binary opposite, i.e., contrastive dual sampling. In addition to data augmentation like other contrastive DCMH methods, we are attempting to construct interventional causality, i.e., twin contrastive interventional-cause learning.

3.5. Interventional-cause learning

As shown in Fig. 6, the paradigm of UDCMH methods (including our method) is the following: input query Q , model M , hash B , and the response R . Besides, we generated S to assist in guiding the model. The degree of correlation between Q and R indicates the performance of M . After pre-sampling by ImgNet and TxtNet, we constructed the interventional-cause learning for UDCMH. Inspired by Structure Causal Model (SCM) (Fan et al., 2022; Pearl et al., 2016, 2000; Wu et al., 2022b), all variables can be assumed to be independent, and intervention operations (Pearl & Mackenzie, 2018) can be introduced for processing. Intervention refers to changing some of the variables while keeping other mechanisms unchanged to explore their inherent invariance principles (Pearl, 1995). Specifically, we extracted the

interventional-cause part: W and N , their relationship is shown in Fig. 6. The detailed relationship is summarized as follows:

- $M \rightarrow B$ and $M \rightarrow S \rightarrow B$. By “causal part”, we mean that B is generated from M by positive learning, and S has a positive effect on B .
- $B \rightarrow R$. As the expected result of model training, we mean that B is the only endogenous element to predict the ground-truth response R .
- $M \xrightarrow{R} W$ and $M \xrightarrow{R} N$. By “interventional-cause part”, we mean that W and N are generated from M by interventional-cause reverse learning.
- $W \nrightarrow R$. As the unexpected result of model training, we can not build the link between W and R .
- $N \nrightarrow B$ and $S \nrightarrow W$. By “interventional-cause part”, there is reverse guidance in the inter subschemes.
- $M \xrightarrow{R} N \rightarrow W$. Through double reverse construction, we assume that N has a positive effect on W .

After reverse construction, model learning can apply spurious correlations between the non-causal part (inter subschemes, e.g., $S \nrightarrow W$) and the ground-truth part (inner subschemes, e.g., $S \rightarrow B$). By assuming $N \rightarrow W$, W and N are confounders between B and S , which opens a backdoor path $M(S) \xrightarrow{R} N \rightarrow W$, thus leading S and W spuriously correlated (Pearl et al., 2016). In addition, there may be interference from specific environmental factors during data collection, resulting in spurious correlations, but they are prone to fluctuations (Arjovsky et al., 2019; Teney et al., 2021).

Our goal is to create an interventional distribution. The non-causal part was formally introduced for collection in W and N , and construct intervention pairs ($N \nrightarrow B$ and $S \nrightarrow W$), which can intervene in the model forward training. Here, we use R_B to represent the recommended response computed by B , i.e.,

$$R_B = \text{Pre}(B), \quad (6)$$

where $\text{Pre}(\cdot)$ represents the final recommendation function. Similarly, the pseudoreturn response R_W generated by W can be represented as follows:

$$R_W = \text{Pre}(W). \quad (7)$$

R_B is the prediction based merely on the causal part B , while R_W measures the predictive power of the intervened part W . Inspired by Cadene et al. (2019), we formulate the joint prediction R under the intervention:

$$R = R_B \odot \text{sigmoid}(R_W), \quad (8)$$

where $\text{sigmoid}(\cdot)$ represents the sigmoid function applied to adjust the output logits of the causal part (i.e., B and S) to compensate for the spurious biases. Different from partitioning the training data, we do not apply supervised modality-specific information but introduce the interventional distribution (Pearl et al., 2016; Tian et al., 2006) to model the CMR tasks. The implementation of causal learning is a thorny issue. Based on the inherent principles of image-text retrieval, we use matching results as prediction R to guide cross-modal learning, and combine the design of formula (5) and formula (8) in specific model training, constructing a contrastive function group to achieve causal intervention. Specifically, on the top of our SCM, we generate interventional distribution by doing interventions W and N , in which fixing B by the specific value S . By stratifying different learning as shown in Fig. 6, we can obtain multiple interventional distributions.

3.6. Twin contrastive learning

The implementation of cross-modal causal intervention learning is hard to ground, and this section will specifically discuss the twin contrastive learning loss function. In Section 3.3, we introduced an objective function formula (5) commonly used in current UDCMH methods.

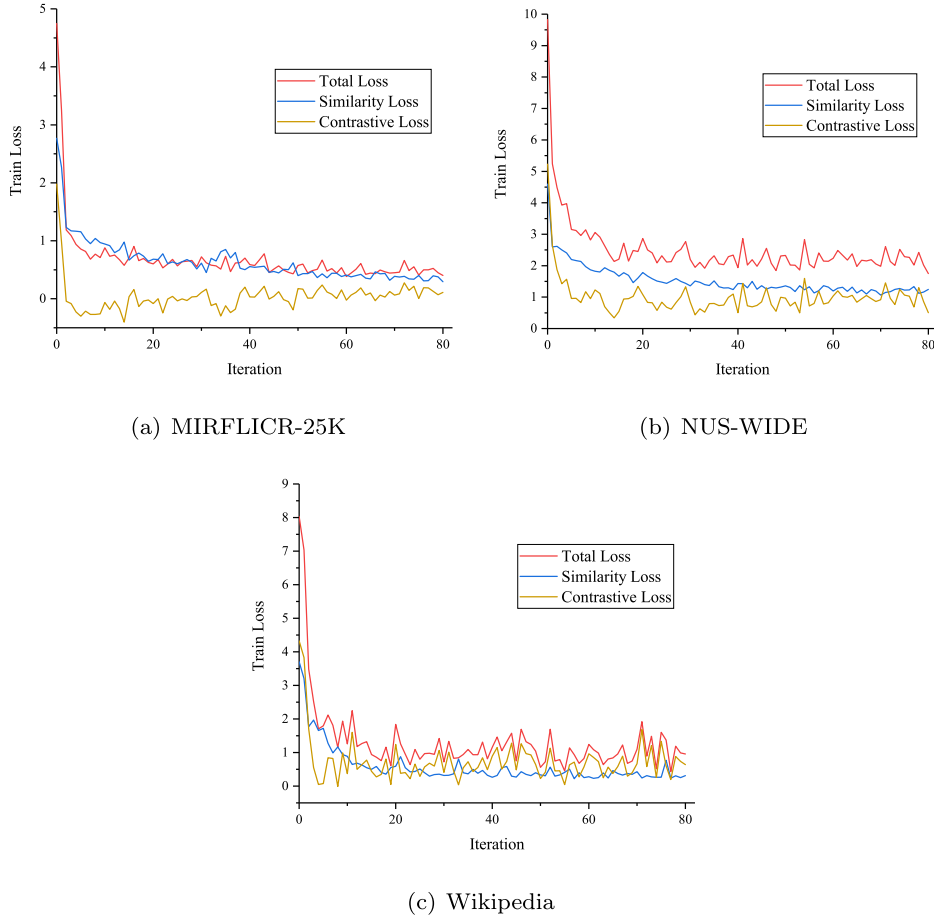


Fig. 11. The convergence curves of TCICH with 128-bits codes length on three datasets.

To standardize the subsequent discussion, we redefine formula (5) as the universal formula:

$$\begin{aligned} \text{loss}(\mathbf{B}, \mathbf{S}) = & \min_{\mathbf{B}_I, \mathbf{B}_T} \|\gamma \mathbf{S} - \cos(\mathbf{B}_I, \mathbf{B}_T)\|_F^2 \\ & + \mu_1 \|\gamma \mathbf{S} - \cos(\mathbf{B}_I, \mathbf{B}_I)\|_F^2 + \mu_2 \|\gamma \mathbf{S} - \cos(\mathbf{B}_T, \mathbf{B}_T)\|_F^2. \end{aligned} \quad (9)$$

The meta loss formula (9) is also used as the loss function of our benchmark module, i.e., similarity loss. For contrastive learning, the loss function in other parts can be on-policy or off-policy. Off-policy settings can reflect the function of different parts but increase the scheme complexity. As this paper is the first attempt to address hash representation and contrastive learning, we simplify scheme design and use on-policy for most settings. In the twin framework, as the semi-processed knowledge generated by reverse construction in Section 3.4, $\mathbf{N}$ can also guide learning on $\mathbf{W}$, so we can obtain the second objective function as follows:

$$\begin{aligned} \text{loss}(\mathbf{W}, \mathbf{N}) = & \min_{\mathbf{W}_I, \mathbf{W}_T} \|\gamma \mathbf{N} - \cos(\mathbf{W}_I, \mathbf{W}_T)\|_F^2 \\ & + \mu_1 \|\gamma \mathbf{N} - \cos(\mathbf{W}_I, \mathbf{W}_I)\|_F^2 + \mu_2 \|\gamma \mathbf{N} - \cos(\mathbf{W}_T, \mathbf{W}_T)\|_F^2. \end{aligned} \quad (10)$$

From Figs. 3 and 5, it can be seen that the farther the original samples are from $\mathbf{W}$ after $\mathbf{N}$ guides, the more beneficial it is for the overall model, so the relationship between $\mathbf{W}$ and $\mathbf{N}$ can also be considered positive for the model learning. So there is positive guidance in the inner subschemes, which is reflected in $\text{loss}(\mathbf{B}, \mathbf{S})$ and $\text{loss}(\mathbf{W}, \mathbf{N})$. Although $\text{loss}(\mathbf{B}, \mathbf{S})$ and $\text{loss}(\mathbf{W}, \mathbf{N})$ are isomorphic, $\mathbf{B}$ and $\mathbf{W}$ are mirrored ($\mathbf{S}$ and $\mathbf{N}$ are also), due to the multi-layer fitting of deep learning frameworks, the matching performance of the overall scheme can also be improved by adding $\text{loss}(\mathbf{W}, \mathbf{N})$.

Unlike the positive guidance of $\mathbf{S}$ on $\mathbf{B}$, as shown in Figs. 3 and 5, because $\mathbf{N}$ is reversely generated by $\mathbf{S}$, the semantic structure relationships contained in $\mathbf{N}$ is unbeneficial to the native hash matrix $\mathbf{B}$. Therefore,

the smaller the loss between $\mathbf{N}$ and $\mathbf{B}$ is, the worse is the performance of overall model, while it can be used as negative validation knowledge. For on-policy, the loss between $\mathbf{B}$ and $\mathbf{N}$ is defined by the standard as follows:

$$\begin{aligned} \text{loss}(\mathbf{B}, \mathbf{N}) = & \min_{\mathbf{B}_I, \mathbf{B}_T} \|\gamma \mathbf{N} - \cos(\mathbf{B}_I, \mathbf{B}_T)\|_F^2 \\ & + \mu_1 \|\gamma \mathbf{N} - \cos(\mathbf{B}_I, \mathbf{B}_I)\|_F^2 + \mu_2 \|\gamma \mathbf{N} - \cos(\mathbf{B}_T, \mathbf{B}_T)\|_F^2. \end{aligned} \quad (11)$$

Similar to the effect of $\mathbf{N}$ on $\mathbf{B}$, $\mathbf{S}$ has the opposite guiding effect on $\mathbf{W}$. The closer the distance between $\mathbf{S}$ and $\mathbf{W}$ is, the worse is the effectiveness of the model, which means that the generated hash codes refer to incorrect information. The loss between $\mathbf{W}$ and $\mathbf{S}$ is defined as:

$$\begin{aligned} \text{loss}(\mathbf{W}, \mathbf{S}) = & \min_{\mathbf{W}_I, \mathbf{W}_T} \|\gamma \mathbf{S} - \cos(\mathbf{W}_I, \mathbf{W}_T)\|_F^2 \\ & + \mu_1 \|\gamma \mathbf{S} - \cos(\mathbf{W}_I, \mathbf{W}_I)\|_F^2 + \mu_2 \|\gamma \mathbf{S} - \cos(\mathbf{W}_T, \mathbf{W}_T)\|_F^2. \end{aligned} \quad (12)$$

As shown in Figs. 3 and 5, in the twin contrastive hashing learning framework, $\text{loss}(\mathbf{B}, \mathbf{S})$ and $\text{loss}(\mathbf{W}, \mathbf{N})$ provide positive guidance for the overall model, while $\text{loss}(\mathbf{B}, \mathbf{N})$ and $\text{loss}(\mathbf{W}, \mathbf{S})$ give negative correction, i.e, there is positive guidance in the inner subschemes, and reverse guidance in the inter subschemes. So, $\text{loss}(\mathbf{B}, \mathbf{S})$ and $\text{loss}(\mathbf{W}, \mathbf{N})$ should be set with positive incentive factors, while $\text{loss}(\mathbf{B}, \mathbf{N})$ and $\text{loss}(\mathbf{W}, \mathbf{S})$ should be given negative corrections. Therefore, the overall loss function L can be expressed as:

$$L = \alpha_1 \text{loss}(\mathbf{B}, \mathbf{S}) - \beta_1 \text{loss}(\mathbf{B}, \mathbf{N}) - \beta_2 \text{loss}(\mathbf{W}, \mathbf{S}) + \alpha_2 \text{loss}(\mathbf{W}, \mathbf{N}), \quad (13)$$

where α_1 , α_2 , β_1 , and β_2 are hyper-parameters. α_1 and α_2 are positive hyper-parameters that measure the weight of positive guidance. β_1 and β_2 are negative hyper-parameters that adjust model convergence speed. On the other hand, as shown in Fig. 1, $\alpha_1 \text{loss}(\mathbf{B}, \mathbf{S})$ refers to the similarity loss of cross-modal semantic matching applied in the baseline module,

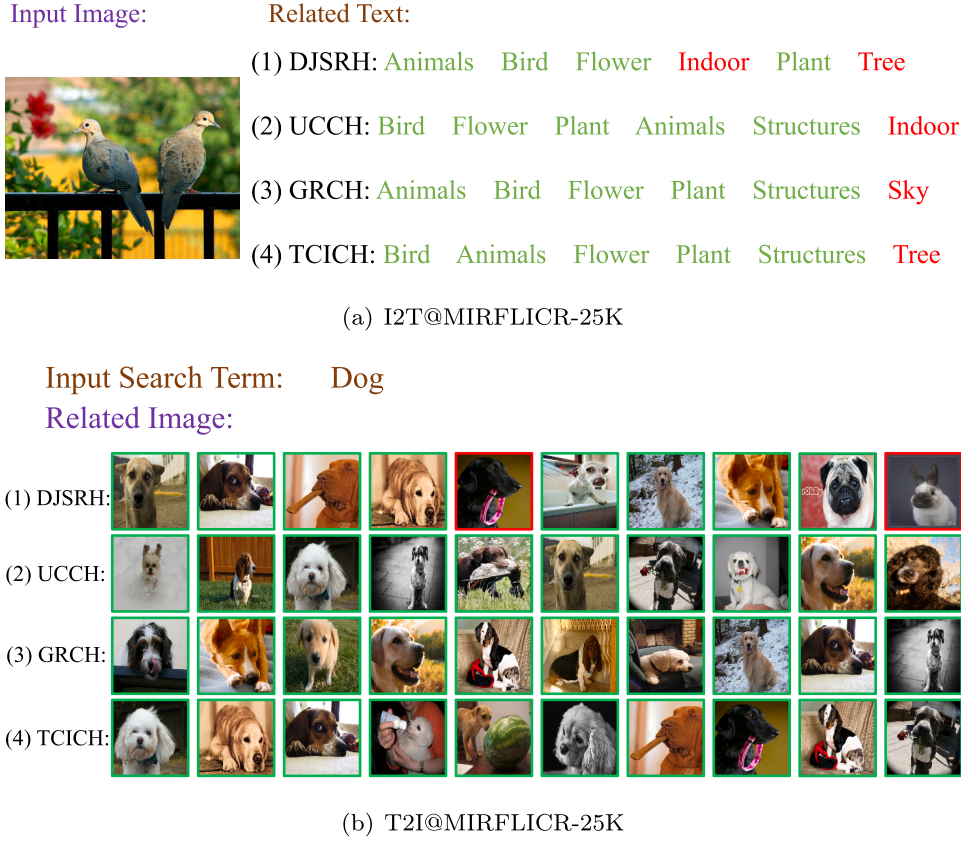


Fig. 12. Comparison of retrieval results with different methods on the MIRFLICR-25K dataset.

while the contrastive loss in this paper is $-\beta_1 \text{loss}(\mathbf{B}, \mathbf{N}) - \beta_2 \text{loss}(\mathbf{W}, \mathbf{S}) + \alpha_2 \text{loss}(\mathbf{W}, \mathbf{N})$. In the training phase, we use interventional construction to utilize four submodules for causal reasoning. In the test phase, we do not use other matrices except for hash matrix $\mathbf{B}$, so our method is an implicit unsupervised method.

Unlike well-known multimodal contrastive models (e.g., CLIP Radford et al., 2021, ALIGN Jia et al., 2021) or causal inference-based multimodal methods, TCICH focuses on introducing self-disturbance during the training process and considers the ambiguity of hash representations, making the overall model more robust. In addition, our scheme is beyond specific multimodal semantic structures, which can align features from different modalities and avoid modality-specific redundancy. This design combines the intrinsic principles of cross-modal hash learning and takes into account existing image-text matching strategies in its implementation, making it highly transferable.

3.7. Optimization

The process of twin contrastive interventional-cause learning is conducted by jointly minimizing the overall loss as formula (13), which can be optimized during batch training by any stochastic gradient descent (SGD) optimization algorithm. In the twin comparative learning framework, the four loss functions learn from each other, making the distance between the original framework and the twin framework as large as possible. The optimization process of our TCICH is summarized in Algorithm 1.

3.8. Out-of-sample extension

In testing and practical scenarios, training-generated TCICH can convert input image samples $\mathbf{i}_x$ to corresponding hash codes, i.e.,

$$\mathbf{B}_I = \tanh(\mathbf{F}_I(\mathbf{i}_x; \Omega)). \quad (14)$$

Algorithm 1: Optimization process of our TCICH.

Input: Database instances $\mathbf{D}$, ImgNet hashing network parameters Ω and TxtNet hashing network parameters Υ , initialization parameters: $\alpha_1, \alpha_2, \beta_1, \beta_2, \delta, \theta, \gamma, \mu_1, \mu_2$, the length of hash codes k .

Output: Overall loss L (minimized), Ω , and Υ .
 Randomly initialize the network parameters Ω and Υ .

```

for iterBatch = 1 to MaxNum  $\tau$  do
  Randomly generate index set  $\phi$ ;
  Extract image and text features to the construct hash matrix
   $\mathbf{B}$  and structural joint semantic similarity metric  $\mathbf{S}$ ;
  Reverse construction: generate twin hash matrix  $\mathbf{W}$  and
  twin similarity metric  $\mathbf{N}$ ;
  Compute forward-propagation;
  Calculate the overall loss  $L$  based on formula (13);
  Update the parameters  $\Omega$  and  $\Upsilon$  by using backpropagation:
   $\Omega = \text{Optimizer-ImgNet.SGD}(\Omega)$ ;
   $\Upsilon = \text{Optimizer-TxtNet.SGD}(\Upsilon)$ .

```

end
return Ω and Υ

Similarly, untrained text instances $\mathbf{t}_x$ can also be converted to

$$\mathbf{B}_T = \tanh(\mathbf{F}_T(\mathbf{t}_x; \Upsilon)). \quad (15)$$

3.9. Computational complexity analysis

In each iteration of the training stage, we construct the structural joint semantic similarity metric $\mathbf{S}$, which costs $O((n+m)(p_I + p_T))$, where n is the size of the dataset, m is the size of random sampling subset, p_I is the feature dimension of ImgNet, and p_T is the feature di-

mension of TxtNet. The computational cost of total model is $O_{TCICH} = O(((n+m)(p_I + p_T) + \Delta)\tau)$, where τ is the iteration time and Δ is the computational complexity of other parts. Because only feature sampling and structural semantic similarity excavation (i.e., Section 3.3) involve matrix convolution and dimension transformation, while the other parts are linear calculations, so $\Delta \ll O((n+m)(p_I + p_T))$. Besides, m , p_I , p_T , and $\tau \ll n$, so O_{TCICH} is linear to the size of datasets n . It is worth noting that our essential design, i.e., twin contrastive learning, does not have excessively high computational complexity, which improves model performance without significantly increasing computational complexity. Therefore, the computational complexity of our TCICH is acceptable for enhancing performance.

3.10. Theoretical analysis

For cross-modal matching tasks, our twin contrastive structures can achieve superior performance than other contrastive schemes, mainly due to the compatibility between twin contrastive structures and hash representations. Because the CMH method requires compressing features into a binary Hamming space, where each bit has two optional values, it is more suitable for coupling the hash space and the contrastive space. Although the hash space can theoretically be further subdivided to accommodate multi-branch contrastive structures, this would actually lose the advantage of binary representation. In addition, we introduced causal intervention in the twin contrastive learning framework to assist in enhancing the matching accuracy, but this is the secondary season.

Objectively speaking, each strategy and method has certain limitations. We have considered many factors when starting this research, but there may still be some constraints unconsidered or unsolved. The potential theoretical boundary conditions of the proposed method include: contrastive learning needs to consider the contrastive differences of the samples, and if the overall sample is evenly distributed in each class, it will weaken the efficiency of the dual contrastive construction. Secondly, because the proposed twin contrastive learning framework overlaps the hash space and the contrastive space, the representation range of the hash space is also potentially related to the number of samples. If the number of samples exceeds the recognition degree, it will also cause a decrease in efficiency.

4. Experimental settings

To validate the performance of TCICH, we conducted extensive experiments on three image-text datasets. In this section, we provide a brief introduction to the datasets, evaluation metrics, and implementation details.

4.1. Datasets

The three experimental datasets are MIRFlickr-25K (Huiskes & Lew, 2008), NUS-WIDE (Chua et al., 2009), and Wikipedia (Pereira et al., 2013). The MIRFlickr-25K (Huiskes & Lew, 2008) dataset contains 25,000 image-text pairs, for a specific image, which may have multiple corresponding texts or zero (nearly 5,000), our experiment mainly used these 20,015 image-text pairs with category information. For model training, we randomly selected 10% of them as the query set and the rest for the retrieval set, from which 5000 image-text pairs were randomly sampled as the training set.

The NUS-WIDE (Chua et al., 2009) dataset collected 269,648 image-text samples from Flickr, which can be divided into 81 classifications, and each image has several semantically related tags. We selected 186,557 image-text cases from the 10 most commonly used categories for main experiments. Among them, 2000 image-text pairs were randomly sampled as the query set, and the remaining pairs were used as the retrieval set, from which 5000 image-text pairs were randomly sampled as the training set.

The Wikipedia (Pereira et al., 2013) dataset contains 2866 images and corresponding documents, and these multi-modal data are divided into 10 categories, each document has over 70 words. In specific experiments, 693 cases were used as the query set, and 2173 cases were used for training. The specific statistics are presented in Table 2.

4.2. Evaluation criterion

To validate the superiority of the proposed TCICH, our experiments use mean Average Precision (mAP) and the precision@topK ($P@topK$) curve as evaluation criteria. The average precision (AP) for the x th query sample is $AP(x) = \frac{1}{Z} \sum_{r=1}^R p(r)rel(r)$, where Z is the number of relevant response, $p(r)$ is the precision of the first r responses. If the r th response is semantically related to the input, then $rel(r) = 1$; otherwise, $rel(r) = 0$. After calculating all AP of input queries $Q = [q_1, q_2, \dots, q_m]$, the mAP is computed by

$$mAP = \frac{1}{m} \sum_{x=1}^m AP. \quad (16)$$

Our experiments are to calculate the average accuracy of all return elements, i.e., $mAP@all$.

mAP is the average precision of a fixed number of return elements. If recording the precision at different numbers of responses, we can draw $P@topK$ precision curve to measure the hash lookup protocol. The $P@topK$ curve is an essential indicator to evaluate the accuracy of the top K responses. For $P@topK$ curve, it is obvious that the larger the region between the curve and the coordinate axis is, the better this method may be.

4.3. Implementation details

In these experiments, we compare 12 SOTA unsupervised cross-modal hash retrieval (UCMH) algorithms, including 3 shallow UCMH methods and 9 UDCMH methods, all the methods used for comparison in this paper have been introduced in the previous Chapter 2. We retrained and retested these UCMH methods with their source codes provided by corresponding authors, or cited the best-recorded value under the same conditions. In model training, for our method and other retrained baselines, image representations are extracted from the AlexNet (Krizhevsky et al., 2017) backbone, and text representations are extracted from the BOW backbone. For shallow CMH methods, we extracted features from ImgNet and TxtNet, and then, these methods directly used extracted features for subsequent training. Because some experimental results are from retrained methods and based on different image sampling baselines, it is acceptable that there were some fluctuations (different from the original paper).

All re-run methods and our method were trained 10 times, then tested separately, the average results were taken as the experimental result. As the precautions to minimize random effects, we also detect changes in the loss function, and if exception occurs in model training, remove this result and retrain again to prevent random risks as much as possible. During the testing phase, on the previously randomly grouped test set, the trained model will generate corresponding hash codes for each sample to be tested, and then will perform cross-model matching to obtain the return result of this sample. As described in Section 4.2, after summarizing the return results of all test samples, $mAP@all$ value and $P@topK$ curve can be calculated.

The model was tested on one NVIDIA GeForce RTX 3090 GPU, which memory size is 24 GB, using PyTorch 1.7.1 + CUDA 11.0 and Python 3.8.13. In the experiments of Section 5.1, our TCICH adopted SGD as the gradient descent strategy, the momentum was 0.9, and the weight attenuation was 0.0005. For all three datasets, $\alpha_1 = 20$, $\alpha_2 = \beta_1 = \beta_2 = 1$, $\delta = 0.4$, and $\gamma = 1.5$. For MIRFlickr-25K, $\theta = 0.9$, $\mu_1 = \mu_2 = 0.1$. For NUS-WIDE, $\theta = 0.6$, $\mu_1 = \mu_2 = 0.1$. For the Wikipedia, $\theta = 0.3$, $\mu_1 = \mu_2 = 0.3$. In addition, when running on MIRFlickr-25K and NUS-WIDE, the learning rate of ImgNet to extract basic features and

Table 2
Basic statistics of the experimental datasets.

Datasets	Image-text pairs	Categories	Training	Query	Retrieval
MIRFlickr-25K (Huiskes & Lew, 2008)	25,000 (20,015 in use)	24	5,000	2,000	18,015
NUS-WIDE (Chua et al., 2009)	269,648 (186,577 in use)	81 (10 in use)	5,000	2,000	184,577
Wikipedia (Pereira et al., 2013)	2,866	10	2,173	693	2,173

spatial features was set to 0.001, while that for Wikipedia was 0.01. The learning rate of TxtNet on three datasets was 0.01. The dataset was trained 10 times in total, and the batch size was 32. The iteration time for MIRFlickr-25K was set to 40, that for Wikipedia was 80, and that for NUS-WIDE was 120.

5. Experimental results and analysis

In this section, we will show the results of experiments along with the corresponding analysis. Concretely, we attempt to answer the following 7 research questions to guide our evaluation objectives in these experiments:

- RQ1: Is the overall performance of our model superior to the SOTA baselines on different datasets? Reply as [Section 5.1](#).
- RQ2: Is the combination of contrastive learning and hashing effective? Reply as [Section 5.2](#).
- RQ3: What is the sensitivity of crucial hyper-parameters? Reply as [Section 5.3](#).
- RQ4: What is the influence of each component for twin contrastive hashing scheme? Reply as [Section 5.4](#).
- RQ5: What are the effects of noise labels and imbalanced data distribution? Reply as [Section 5.5](#).
- RQ6: What is the training and testing cost of twin contrastive hashing scheme? Reply as [Section 5.6](#).
- RQ7: What is the convergence of the twin contrastive loss? Reply as [Section 5.7](#).

After analyzing these 7 research questions, we summarized the advantages and disadvantages of TCICH in [Section 5.8](#).

5.1. Performance comparison

In this section, we compared TCICH with most SOTA UCMH methods. For the two evaluation criteria, the results of $mAP@all$ are presented in [Tables 3, 4, and 5](#), the $P@topK$ curves are shown in [Fig. 7](#). To ensure statistical significance, we use confidence intervals as variance to support the effectiveness of our approach. Compared with other UCMH methods, TCICH performs better on most parts of $mAP@all$. The performance of TCICH decreases slightly when short hash coding (16-bit), but this is due to the characteristics of contrastive learning and hash space. We will conduct the comprehensive analysis in [Section 5.2](#). Then, we measured $P@topK$ curves on three datasets (except GCRH [Wei et al. \(2025\)](#) is 64-bit, due to which original work no relevant data), as shown in [Fig. 7](#). According to the downward trends of the curves, TCICH has slight improvements compared with other methods. Only on text-to-image@MIRFLICKR-25K curves, GCRH has slight advantages, but this also reflects the global impact of integrating spatial contrast into the model, which is similar to our idea. On the basis of these indicators, TCICH significantly improved compared to other unsupervised methods.

To further analyze innovation, we conducted in-depth comparisons of four representative methods with our TCICH. First, as mentioned earlier, considering generalization and fairness, we adopted a classical loss function for each submodule, i.e., formula (5). The overall objective function of DJSRH ([Su et al., 2019](#)) is also formula (5), which is used in

many UDCMH methods. Compared with DJSRH, from 32-bit to 128-bit, our method achieved obvious improvements in cross-modal matching on three different datasets. Performance differences between TCICH and DJSRH clearly show that reverse construction has significant improvements in various aspects, indicating that our twin contrastive hashing learning framework is simple but efficient. It is worth noting that our approach of combining contrastive learning with hashing is for the overall framework and is not limited to a specific objective function, which can be easily extended to other methods. Second, although both our proposed TCICH and UCCH ([Hu et al., 2022](#)) apply contrastive learning to CMR tasks, different design ideas also lead to differences in experimental performance. UCCH discussed the influence of false-negative pairs (FNPs) and proposed the cross-modal ranking learning loss (CRL) for instance training, which only focuses on the details and neglects the overall scheme design. Different from UCCH, we analyzed the hash space and Hamming distance from the overall design perspective and constructed a twin hashing framework to achieve contrastive learning. It is obvious that overall model optimization is superior to bit-by-bit optimization. Third, SACH ([Cui et al., 2024](#)) has similar results to our proposed method in many indicators by concurrently employing both high-dimensional representations and discriminative representations to construct a semantic correlative matrix. For contrastive learning, SACH introduced decoupled contrastive learning (DCL) ([Yeh et al., 2022](#)) to construct high-order semantic alignment, while this design has already been used in other methods and SACH does not have further specific improvement. Besides, this application is effective, it may lack improvements in the underlying principles of contrastive learning and is not fundamentally different from other contrastive CMH methods. Both considering performance and innovation, although SACH performs similarly to our proposed method in experimental performance, our improvements are more novel and have greater significance for CMR tasks. Fourth, similar to our approach proposed from the global perspective, GCRH ([Wei et al., 2025](#)) integrated graph convolutional networks and contrastive learning into the whole scheme design. This macro improvement is effective and even performs best on some indicators. Based on the performances of GCRH and TCICH, it can be concluded that this global-level design is effective, stable, and easy to migrate. However, compared to GCRH, we also considered hashing in the model construction and attempted to introduce causal learning. The more comprehensive design makes our method perform better overall.

By analyzing UCCH, SACH and GCRH, many existing contrastive CMH methods only introduce contrastive learning in model design to achieve data augmentation, which can improve matching performance, but with little or no proprietary innovation for cross-modal hashing. This also makes these contrastive methods essentially similar to non-contrastive methods. Unlike existing comparative CMH methods, we construct the model design from a global perspective while considering the characteristics of contrastive learning and hashing meantime. Our twin contrastive learning approach emphasized the intrinsic principle from the above factors and underscored essential innovation to enhance matching performance. In addition, there is a little decrease in the performance of TCICH in short encoding (16-bit), which is acceptable because practical applications focus more on longer encoding, we will also conduct a supplementary analysis in [Section 5.2](#). As shown in the experimental results, it can be seen that our method saves training costs but improves performance, proving that our twin contrastive learning is more effective for achieving CMR tasks.

Table 3

The $mAP@all$ results of different hash code lengths on MIRFLICKR-25K datasets. The best results are highlighted in bold.

Method		Image to Text				Text to Image			
		16-bits	32-bits	64-bits	128-bits	16-bits	32-bits	64-bits	128-bits
CVH (Kumar & Udupa, 2011)	2011IJCAI	0.610	0.601	0.591	0.584	0.618	0.610	0.597	0.591
IMH (Song et al., 2013)	2013SIGMOD	0.600	0.595	0.588	0.576	0.595	0.594	0.592	0.592
CMFH (Ding et al., 2014)	2014CVPR	0.546	0.547	0.546	0.548	0.541	0.542	0.542	0.543
UGACH (Zhang et al., 2018)	2018AAAI	0.680	0.691	0.703	0.702	0.671	0.679	0.687	0.690
DJSRH (Su et al., 2019)	2019ICCV	0.655	0.657	0.671	0.680	0.665	0.660	0.662	0.683
JDSH (Liu et al., 2020)	2020SIGIR	0.680	0.686	0.700	0.698	0.679	0.692	0.696	0.700
DGCPN (Yu et al., 2021)	2021AAAI	0.708	0.716	0.723	0.730	0.702	0.710	0.715	0.720
HNH (Zhang et al., 2021)	2021WWW	0.691	0.690	0.701	0.693	0.697	0.692	0.695	0.716
CIRH (Zhu et al., 2022)	2022TKDE	0.697	0.704	0.717	0.726	0.699	0.712	0.722	0.735
UCCH (Hu et al., 2022)	2022TPAMI	0.680	0.681	0.682	0.684	0.678	0.679	0.681	0.686
SACH (Cui et al., 2024)	2024NN	0.724	0.726	0.732	0.734	0.727	0.736	0.740	0.743
GCRH (Wei et al., 2025)	2025DSE	0.713	0.725	0.734	–	0.730	0.739	0.743	–
TCICH		0.698	0.728	0.736	0.739	0.701	0.741	0.746	0.751

Table 4

The $mAP@all$ results of different hash code lengths on NUS-WIDE datasets. The best results are highlighted in bold.

Method		Image to Text				Text to Image			
		16-bits	32-bits	64-bits	128-bits	16-bits	32-bits	64-bits	128-bits
CVH (Kumar & Udupa, 2011)	2011IJCAI	0.458	0.432	0.410	0.392	0.474	0.445	0.419	0.398
IMH (Song et al., 2013)	2013SIGMOD	0.401	0.406	0.410	0.421	0.370	0.375	0.386	0.398
CMFH (Ding et al., 2014)	2014CVPR	0.340	0.341	0.347	0.350	0.323	0.316	0.317	0.312
UGACH (Zhang et al., 2018)	2018AAAI	0.543	0.580	0.577	0.603	0.559	0.591	0.583	0.614
DJSRH (Su et al., 2019)	2019ICCV	0.503	0.540	0.561	0.570	0.512	0.524	0.538	0.555
JDSH (Liu et al., 2020)	2020SIGIR	0.506	0.539	0.564	0.571	0.521	0.534	0.540	0.560
DGCPN (Yu et al., 2021)	2021AAAI	0.505	0.515	0.554	0.568	0.520	0.535	0.544	0.560
HNH (Zhang et al., 2021)	2021WWW	0.558	0.589	0.586	0.598	0.587	0.610	0.612	0.615
CIRH (Zhu et al., 2022)	2022TKDE	0.573	0.602	0.606	0.623	0.579	0.607	0.614	0.628
UCCH (Hu et al., 2022)	2022TPAMI	0.518	0.543	0.568	0.574	0.525	0.549	0.570	0.583
SACH (Cui et al., 2024)	2024NN	0.612	0.624	0.627	0.633	0.614	0.621	0.625	0.630
GCRH (Wei et al., 2025)	2025DSE	0.615	0.622	0.638	–	0.621	0.635	0.652	–
TCICH		0.598	0.626	0.640	0.646	0.600	0.640	0.655	0.663

Table 5

The $mAP@all$ results of different hash code lengths on Wikipedia datasets. The best results are highlighted in bold.

Method		Image to Text				Text to Image			
		16-bits	32-bits	64-bits	128-bits	16-bits	32-bits	64-bits	128-bits
CVH (Kumar & Udupa, 2011)	2011IJCAI	0.146	0.149	0.155	0.156	0.275	0.236	0.186	0.179
IMH (Song et al., 2013)	2013SIGMOD	0.157	0.148	0.136	0.134	0.239	0.238	0.240	0.241
CMFH (Ding et al., 2014)	2014CVPR	0.163	0.176	0.163	0.173	0.163	0.176	0.163	0.175
UGACH (Zhang et al., 2018)	2018AAAI	0.388	0.395	0.401	0.403	0.401	0.445	0.486	0.509
DJSRH (Su et al., 2019)	2019ICCV	0.355	0.381	0.396	0.401	0.393	0.441	0.475	0.491
JDSH (Liu et al., 2020)	2020SIGIR	0.351	0.383	0.399	0.406	0.398	0.448	0.480	0.492
DGCPN (Yu et al., 2021)	2021AAAI	0.394	0.401	0.409	0.410	0.516	0.531	0.537	0.543
HNH (Zhang et al., 2021)	2021WWW	0.368	0.392	0.401	0.409	0.480	0.511	0.530	0.543
CIRH (Zhu et al., 2022)	2022TKDE	0.375	0.399	0.406	0.411	0.478	0.513	0.529	0.543
UCCH (Hu et al., 2022)	2022TPAMI	0.384	0.395	0.410	0.414	0.486	0.509	0.526	0.544
SACH (Cui et al., 2024)	2024NN	0.395	0.403	0.411	–	0.512	0.530	0.539	–
GCRH (Wei et al., 2025)	2025DSE	0.403	0.412	0.419	–	0.522	0.541	0.552	–
TCICH		0.382	0.413	0.420	0.426	0.496	0.520	0.553	0.558

5.2. Contrastive learning and hashing

Introducing hash technology is to reduce computational complexity and storage space, so the hash encoding length in most methods does not exceed 128 bits. However, we introduced in previous chapters that owing to the binary representation of hash codes, the key point of this paper is combining contrastive learning with hashing, and we use hash space to achieve contrastive learning. Contrastive learning emphasizes the contrast of different samples, but if the hash space is too small and the distance of different samples may be too near, it will lead to insufficient discrimination of dissimilar instances and limit the performance of contrastive learning. To verify the impact of hash space and contrastive

learning on TCICH model performance, we conducted three additional experiments on MIRFlicker-25K datasets.

First, we compared it with another contrastive UDCMH SOTA baseline, i.e., UCCH (Hu et al., 2022) in long hash code lengths include 96, 192, and 256 bits. The variant mAP results are shown in Table 6. By combining Tables 3 and 6, it can be seen that the overall performance of UCCH does not increase significantly when coding is longer. This is because UCCH established contrastive label ranking to enhance semantic alignment, but did not consider the characteristics of the hash space. Unlike UCCH, we integrate both contrastive learning and hashing in the scheme construction, considering not only data augmentation but also whole model design. The performance of TCICH improves significantly

Table 6The $mAP@all$ results of long hash codes on MIRFlickr-25K datasets.

Method	Image to Text			Text to Image		
	96-bits	192-bits	256-bits	96-bits	192-bits	256-bits
UCCH (Hu et al., 2022)	0.688	0.691	0.689	0.689	0.692	0.690
TCICH	0.745	0.750	0.757	0.755	0.761	0.763

as the hash space increases, twin contrastive learning has better discriminative ability for samples with obvious differences. In summary, the whole is greater than the sum of the parts, our TCICH is novel and effective.

Second, we compared the model performance with different hashing lengths on the MIRFlickr-25K datasets, including seven variants, from 16 to 256 bits. As the hashing bit increases, long binary codes can represent more different classes and the maximum Hamming distance also increases, which is more conducive to the availability of contrastive learning. As shown in Fig. 8, the model performance significantly decreases when the length is 16 and 32 bits, small hash space would reduce the variability of contrastive samples. Choosing appropriate code lengths will be most beneficial for the combination of contrastive learning and hashing.

To make our method more practical for deployment, we further discussed the adaptability of TCICH in short hash-length scenarios. As mentioned in Section 3.1, hash representation can involve semantic compression and reduce training expenses, e.g., compressing 4096-dimension to 128-dimension. Shorter hash codes may lose some information during the hash generation process, resulting in undesirable input of subsequent feature learning. Based on our other analyses, we assume that there is an empirical correlation between contrastive learning, hash space, and batch size. Therefore, we adjusted the size of batch training and supplemented some experiments on MIRFlickr-25K datasets with short hash codes, as shown in Fig. 9. For the MIRFlickr-25K dataset, when the batch size is small (e.g., 8, 16, 24), model precisions at all hash lengths are unsatisfactory due to poor spatial contrast and inappropriate learning rates. By adjusting the settings, the model at most hash codes (32 bits or more) can achieve efficient performance when the batch size is 32, while 16-bit hash codes perform best when the batch size is around 64. Subsequently, excessively large and nonempirical batch sizes would lead to decreases in average accuracy for all hash lengths. These detail differences are also obtained in the NUS-WIDE dataset. For specific implementation of twin contrastive hash learning, suitable batch sizes can appropriately improve matching accuracy when using short hash lengths, which can adapt to real-world applications where computational efficiency is a priority.

5.3. Parameters study

In our TCICH model, nine types of parameters must be adjusted, i.e., α_1 , α_2 , β_1 , and β_2 in formula (13), δ , θ , γ , μ_1 , and μ_2 in formula (5). It is worth noting that our approach of combining contrastive learning with hashing is for the overall framework and is not limited to a specific objective function. The performance of hyper-parameters in formula (5) is not the focus of our research, and some other works (Su et al., 2019) have fully discussed formula (5), which can converge to the optimum. Secondly, our twin contrastive hashing framework includes multiple modules, and after multiple deep learning training, the role of hyper-parameters within the submodules maybe masked by the hyper-parameters of the overall module. Thirdly, because a single formula (5) can converge, the discussion of these intra-module hyper-parameters can be regarded as the convergence analysis. More importantly, in the specific implementation of twin contrastive learning, for the previously introduced similarity loss and contrastive loss, which are transformed into settings of formula (13), i.e., similarity loss is $\alpha_1 \text{loss}(\mathbf{B}, \mathbf{S})$, and contrastive loss is $-\beta_1 \text{loss}(\mathbf{B}, \mathbf{N}) - \beta_2 \text{loss}(\mathbf{W}, \mathbf{S}) + \alpha_2 \text{loss}(\mathbf{W}, \mathbf{N})$. For these

reasons, we omitted these unnecessary discussions and focused on discussing α_1 , α_2 , β_1 , and β_2 in formula (13), as shown in Table 7, on MIRFlickr-25K datasets when hash coding is 128-bit.

To increase the robustness of the claims, we designed an improved grid search method to achieve the sensitivity analysis of hyper-parameters. Traditional grid search uses exhaustive search to explore the effects of all potential values. However, traversing and adjusting 4 parameters one by one seems too cumbersome. Meanwhile, unlike normal parameter tuning, our design incorporates some prior knowledge to facilitate a more rational parameter search process and achieve improved grid search. Our improved grid search algorithm for parameter tuning is summarized in Algorithm 2.

Algorithm 2: An improved grid search algorithm for parameter tuning.

Input: Hyper-parameters: α_1 , α_2 , β_1 , and β_2 in formula (13).
Output: Optimized values of α_1 , α_2 , β_1 , and β_2 .
 Scope setting: $\alpha_1 \in [0, 100]$, $\alpha_2, \beta_1, \beta_2 \in [0, 10]$, which all are integers.
for $\alpha_1 = 0$ **to** 100 **do**
 Set the new value of α_1 : (1) Try extreme values to find the most suitable range, step=20; (2) Fine-tune α_1 within the empirical area, step=1;
 Based on α_1 , traverse other parameters, step=1;
 Compute model performance and record;
 Select the optimal parameters;
end
return α_1 , α_2 , β_1 , and β_2

In the twin contrastive hashing framework, $\text{loss}(\mathbf{B}, \mathbf{S})$ and $\text{loss}(\mathbf{W}, \mathbf{N})$ provide positive guidance to the model, while $\text{loss}(\mathbf{B}, \mathbf{N})$ and $\text{loss}(\mathbf{W}, \mathbf{S})$ serve as negative knowledge to assist in supervising. When $\alpha_1 = 1$, $\alpha_2 = \beta_1 = \beta_2 = 0$, this overall loss function $L = \text{loss}(\mathbf{B}, \mathbf{S})$, which is also the objective function for many unsupervised baselines (Su et al., 2019), indicates that the model performance is relatively average. When other losses are introduced into the model, the effect of contrastive learning is reflected and the performance is improved. In the improved grid search of 4 parameters tuning, we first explored the boundaries of different parameters and then fine-tuned them to determine the optimal value. For α_1 , it is worth noting that the learning of the model should still focus on positive knowledge, so we keep other values unchanged and gradually increase the value of α_1 to increase the effect of positive learning, i.e., 5, 10, and 20. During this process, the performance of the model has improved, indicating that we should set higher weights for positive guidance than for negative correction. As α_1 continues to increase, such as 100, excessive positive effects will result in almost no effect of negative knowledge, and the performance of the model will decrease again, approaching the original model, which means that negative contrast knowledge is also indispensable. For other parameters, i.e., α_2 , β_1 , and β_2 , excessive weight does not positively motivate the model (columns 4–6 and 8–10 on the right). Although we directly inverted $\mathbf{W}$ and $\mathbf{N}$ when constructing $\mathbf{B}$ and $\mathbf{S}$ in reverse, but fixing other values and increasing α_2 do not have the good effect, which means that twin learning knowledge, even if $\text{loss}(\mathbf{W}, \mathbf{N})$ is positive for model, can only serve as auxiliary information. Through these systematic explorations of hyper-parameter tuning, each sub-module can be set by empirical configuration.

5.4. Ablation study

On the MIRFlickr-25K datasets, we designed different ablation experiments to analyze the performance of different components of TCICH. These variants run for 40 epochs and hash length of 128 bits. Ablation study results are shown in Table 8. Our design fully considers two modal semantic alignment, which are implemented as shown in formula

Table 7The $mAP@all$ effects of the parameters in formula (13) (part).

	α_1	1	1	1	1	1	5	1	1	1	10	1	1	1	20	50	100
	α_2	0	1	0	0	1	1	5	1	1	1	10	1	1	1	1	1
<i>Parameters</i>	β_1	0	0	1	0	1	1	1	5	1	1	1	10	1	1	1	1
	β_2	0	0	0	1	1	1	1	1	5	1	1	1	10	1	1	1
$mAP@all$	I2T	0.668	0.585	0.589	0.701	0.589	0.719	0.568	0.570	0.580	0.720	0.567	0.567	0.588	0.739	0.727	0.583
	T2I	0.684	0.585	0.590	0.699	0.589	0.721	0.572	0.571	0.580	0.728	0.570	0.571	0.588	0.751	0.731	0.608

Table 8The $mAP@all$ on MIRFlickr-25K to evaluate the effectiveness of each component in our TCICH.

Method	Configuration	I2T	T2I
TCICH-1	$\mathcal{S} = \mathcal{S}_I$ in formula (4)	0.697	0.692
TCICH-2	$\mathcal{S} = \mathcal{S}_T$ in formula (4)	0.682	0.682
TCICH-3	Change $\tanh(\cdot)$ to $\text{sign}(\cdot)$ in hash generation	0.685	0.685
TCICH-4	No contrastive dual sampling	0.684	0.690
TCICH-5	No interventional-cause learning	0.691	0.697
TCICH-6	Direct removal of confounding edges	0.696	0.684
TCICH-7	Introduce graph-neighbor coherence preserving networks	0.740	0.745
TCICH-8	$\mathbf{N} = (\mathbf{E} - \mathbf{S})/2$	0.701	0.732
TCICH-9	$\mathbf{W} = (\mathbf{E} - \mathbf{B})/2$	0.707	0.733
TCICH-10	$\mathbf{N} = (\mathbf{E} - \mathbf{S})/2$ and $\mathbf{W} = (\mathbf{E} - \mathbf{B})/2$	0.725	0.747
TCICH		0.739	0.751

(4), and we first discuss the impact of removing individual modalities (image or text). TCICH-1 only uses the $\mathcal{S}_I$ as the structural joint semantic similarity metric $\mathcal{S}$, while TCICH-2 only uses the $\mathcal{S}_T$ as the $\mathcal{S}$. The two experiments prove that the effective construction of $\mathcal{S}$ is essential for the model. TCICH-3 uses $\text{sign}(\cdot)$ instead of $\tanh(\cdot)$ in hash generation. The use of adaptive learning $\tanh(\cdot)$ can effectively improve the experimental performance. To discuss the progressiveness of the proposed causal learning scheme, we set up different variants for analysis: TCICH-4 applied standard contrastive sampling instead of contrastive dual sampling; TCICH-5 only implemented data augmentation and no interventional causality in causal subsequent learning; TCICH-6 excluded soft interference and adopted direct removal of confounding edges to achieve causal intervention, while all these variants showed performance degradation. For analysis of the scalability and exploring other similarity metrics, we referred to DGCPN (Yu et al., 2021) and introduced graph-neighbor coherence preserving networks to measure alignment, rather than auxiliary similarity matrices, i.e., TCICH-7. From the experimental results, it can be seen that adding other semantic alignment methods is also beneficial for the overall model, but it is relatively limited, which indicates that our solution is stable and not limited to specific alignment strategies. As mentioned earlier, reverse construction can be performed in many ways. In the ablation experiments TCICH-8, TCICH-9, and TCICH-10, we subtracted $\mathbf{B}$ and $\mathbf{S}$ using the unit matrix and then normalized to $\{-1, +1\}$, instead of negation. These three variants have their own strengths in terms of performance compared with our original TCICH, which indicates that the twin contrastive hashing framework is effective and easy to expand, not limited by differences in specific construction details. These ablation experiments explore component design from multiple aspects. The core innovation of our twin contrastive learning framework is combine contrastive learning with hash representation, which makes the train-generated model less susceptible to environment-specific perturbation and enhances matching accuracy.

5.5. Verification for beyond-scenarios

In this section, we validate some issues in other scenarios, namely noise-robust retrieval, retrieval for rare classes, and standard deviations. Firstly, cross-modal retrieval often grapples with challenges arising from noisy data, referring to the settings of other works (Jiang & Li, 2025; Zhang et al., 2022), we conducted validation with noisy labels. The noise-robust retrieval experiments are on MIRFlickr-25K datasets, hash

Table 9

Noise-robust cross-modal retrieval on MIRFlickr-25K datasets.

Task	No noise		20% noisy labels		40% noisy labels	
	64-bits	128-bits	64-bits	128-bits	64-bits	128-bits
Image to Text	0.736	0.739	0.695	0.701	0.648	0.652
Text to Image	0.746	0.751	0.702	0.713	0.651	0.655

codes with 64-bits and 128-bits, ratios of noisy labels (i.e., RON) set as 20% and 40%. Different RON values represent different levels of noisy labels, such as 20% means that the input contains 20% missing or wrong labels. The experimental results are shown in Table 9. Due to the introduction of self-disturbance in twin contrastive learning, TCICH has good immunity to noise interference.

Secondly, long-tailed category distributions are common in multi-modal datasets. For example, NUS-WIDE has a total of 269,648 samples with 81 labels, but there are 186,557 samples that contain the top 10 labels, 195,834 samples that contain the top 21 labels, and only 73,814 samples that do not contain the top 21 labels. To compare imbalance or rare class distributions, we conducted additional experiments on NUS-WIDE-21 in addition to the previous experiments on NUS-WIDE-10, as shown in Table 10. For the supplementary experiments on the NUS-WIDE-21, we set the retrieval set as 193,834 instances, and other settings same as NUS-WIDE-10. Obviously, due to an imbalance in semantic distribution, it does hurt matching performance; however, twin contrastive learning can alleviate this issue by reverse construction and has not shown significant fluctuations.

Thirdly, we conducted experiments using multiple different random seeds and calculated the standard deviation of the results. Specifically, on MIRFlickr-25K, NUS-WIDE, and Wikipedia datasets, analyze the standard deviation of the loss function training at stages 10, 20, 40, and 80 epochs. The relevant results at 128-bit hash codes are shown in Table 11. From the experimental results, it can be seen that TCICH may experience some fluctuations at the beginning and then tend to stabilize. These findings demonstrate that our method exhibits strong robustness and stability across different datasets and random seed settings.

5.6. Training and testing efficiency

The training and testing efficiency experiment was conducted using the MIRFlickr-25K datasets when hash coding is 64-bits. We compared the efficiency of TCICH with that of three unsupervised SOTA CMH methods, i.e., DJSRH (Su et al., 2019), UCCH (Hu et al., 2022), and GCRH (Wei et al., 2025). Table 12 shows a comparison of three aspects: training time, hash encoding time, and testing time. The training time includes 5 epochs, 10 epochs, and 20 epochs; Hash time includes image generation and text generation. The batch processing settings are shown in Table 2.

For training time and image hash encoding time, although contrastive learning was applied in UCCH, GCRH, and TCICH, different contrastive learning designs lead to different time requirements. UCCH and GCRH focus on entire instances; the essence of TCICH is hash representation contrastive learning. As a classic unsupervised method, the per-cycle epoch of DJSRH is shorter, but at a disadvantage in other indicators. Text generation has similar time consumption due to the text processing in current methods being analogous, and for testing time, it

Table 10

The analysis of imbalance or rare class distributions on NUS-WIDE-10 and NUS-WIDE-21 datasets.

Dataset	Image to Text				Text to Image			
	16-bits	32-bits	64-bits	128-bits	16-bits	32-bits	64-bits	128-bits
NUS-WIDE-10	0.598	0.626	0.640	0.646	0.600	0.640	0.655	0.663
NUS-WIDE-21	0.556	0.617	0.630	0.634	0.567	0.625	0.633	0.641

Table 11The standard deviation of mAP results on MIRFlickr-25K, NUS-WIDE, and Wikipedia datasets with 128-bits hash codes.

Dataset	Image to Text				Text to Image			
	10 epochs	20 epochs	40 epochs	80 epochs	10 epochs	20 epochs	40 epochs	80 epochs
MIRFlickr-25K	0.0040	0.0031	0.0025	0.0025	0.0043	0.0032	0.0027	0.0026
NUS-WIDE	0.0034	0.0032	0.0027	0.0024	0.0029	0.0027	0.0024	0.0025
Wikipedia	0.0071	0.063	0.0055	0.0045	0.064	0.0055	0.0051	0.0040

Table 12

Training and testing efficiency analysis on MIRFlickr-25K datasets when hash coding is 64-bits.

Method	Training time (s)			Hash encoding time (ms)		Testing time (s)
	5 epochs	10 epochs	20 epochs	Image	Text	
DJSRH (Su et al., 2019)	39.49	79.21	155.83	24.70	3.40	19.87
UCCH (Hu et al., 2022)	143.74	307.19	592.62	50.32	3.41	24.51
GCRH (Wei et al., 2025)	160.85	331.06	653.71	129.02	3.40	25.96
TCICH	51.57	104.31	210.20	25.01	3.41	21.36

Table 13

Training time and sample loading time on different datasets.

Dataset	1 epoch@64-bits (s)	1 epoch@128-bits (s)	Sample loading time
MIRFlickr-25K	10.38	10.54	26.88
NUS-WIDE	10.08	10.03	45.81
Wikipedia	7.59	7.68	3.55

mainly involves hash encoding generation, so there is also no significant difference. Compared with training space, UCCH and GCRH also occupied more GPU memory during the training process, because bit-by-bit optimization required a large matrix calculation, which is much greater than the consumption of DJSRH and TCICH (TCICH is close to DJSRH). During the entire training and testing phase, the GPU space occupied by TCICH does not exceed 12GB.

In addition, we supplemented the discussion on training time with the size of the training set, as shown in Table 13. Because our experiment used batch training and the batch sizes of the MIRFlickr-25K and NUS-WIDE datasets were the same, the training time of one epoch was similar, while on Wikipedia is slightly smaller. We also added extra time statistics for sample import, which is positively correlated with the size of the dataset.

To further evaluate the complexity of TCICH with respect to training scale, we conduct different training sets by selecting various numbers of instances from MIRFlickr-25K dataset, the hash length is set to 128 bits. Fig. 10 plots the training time of TCICH on these training sets, which clearly indicates that the computational cost of TCICH is basically linear with the training scale. As the training set approaches the entire dataset, there is a significant increase in training time; however, this is primarily limited by computer hardware. Considering that it is rare to learn all instances at once, the overall computer complexity is still kept at $O(n)$.

5.7. Convergence analysis

As mentioned earlier, in the specific implementation of the proposed strategy, the introduced similarity loss and contrastive loss are included in the loss function formula (13), in which similarity loss is the first term and contrastive loss is the other term, i.e., similarity loss is $\alpha_1 \text{loss}(\mathbf{B}, \mathbf{S})$, and contrastive loss is $-\beta_1 \text{loss}(\mathbf{B}, \mathbf{N}) - \beta_2 \text{loss}(\mathbf{W}, \mathbf{S}) +$

$\alpha_2 \text{loss}(\mathbf{W}, \mathbf{N})$. In addition to analyzing the effects of different parameters in Section 5.3, we further analyze the convergence of the loss function on three datasets. The experimental results are shown in Fig. 11, where the hash code is 128 bits. The objective values of TCICH decrease quickly and reach stable states after a few epochs, while having some oscillations. Based on the previous introduction, these two loss functions play different roles in the model. Similarity loss is mainly used for semantic alignment to achieve cross-modal matching, and contrastive loss can enhance the similarity loss and improve the matching performance. In the overall loss design, similar losses have a greater weight and are generally consistent with the downward trend of the overall loss. Meanwhile, since we apply the causal intervention strategy to construct contrastive loss, the causal interference weight is not too large.

5.8. Advantages v.s. Disadvantages

Through the discussion and experiments in the previous sections, we analyzed the correlation between contrastive learning with spatial range in detail, and utilized the binary antonym of hash space to construct contrastive matrices in model learning. Then we will proceed with a conclusive theoretical analysis here. For innovation, we are the first to combine contrastive learning with hash space; For experiments, our design is effective compared with other methods. In addition to data augmentation like other contrastive DCMH methods, we are attempting to construct interventional causality. As a first attempt, our method still has some limitations: (1) Contrastive learning requires sample difference, learning performance is better in extremely different samples (with a large Hamming distance), and return-all retrieval results may limit model ability. Therefore, our method has a relatively mediocre retrieval performance for return-all matching results and only has a weak advantage compared with other methods. (2) We believe that an essential factor to consider when combining contrastive learning and hashing is the hash space size (hash code length), i.e., the relationship between the hash space and the batch size. When the hash space matches the sampling size, that is, batch learning at an appropriate scale can also alleviate the impact of hash space size with sample differences, and enhance the applicability of our model in practical scenarios where the hash code length or sample diversity is constrained. (3) Owing to the design of our twin contrastive learning framework, the model can achieve

better results. However, after reaching the optimal performance, twin contrastive frameworks may be still learning and the loss function may be in the oscillation, which can cause additional training costs. In the future, we are planning to conduct follow-up research on these issues.

5.9. Cross-modal retrieval instances

Fig. 12 displays some cross-modal retrieval instances of our method and three representative UDCMH methods (DJSRH Su et al., 2019, UCCH Hu et al., 2022, and GCRH Wei et al., 2025) on MIRFLICR-25K datasets. From the retrieval examples, it can be seen that existing deep methods have good matching accuracy for these semantically clear samples, while our TCICH performs well in overall performance. For a specific input term, our TCICH can obtain correct responses (high matching degree) from positive guidance, while mismatched responses can be excluded by reverse guidance. Therefore, TCICH can be adapted to actual CMR tasks.

6. Conclusions

To provide more reliable knowledge guidance for UDCMH methods and effectively leverage the function of contrastive learning, we designed twin contrastive interventional-cause hashing for unsupervised cross-modal retrieval (TCICH). Unlike other contrastive DCMH methods that only use contrastive learning to analyze instance features, we are the first to apply contrastive learning in model design, not limited to data augmentation. Second, we proposed a new strategy for constructing an intrinsically interpretable UDCMH model that conducts interventions in the training process to create multiple interventional knowledge. Then, TCICH approached the causal reasoning invariant across different knowledge while filtering out the unstable spurious response. Extensive experiments were conducted on three datasets, and the results demonstrated the effectiveness of the proposed scheme. It is worth noting that we are the first work to apply contrastive learning and interventional-cause learning with hash space, so this design can also be referenced by other fields. Meantime, we also plan to continue in-depth research on some limitations of our proposed method and explore multi-level contrastive mechanisms in the future.

CRedit authorship contribution statement

Bo Li: Writing – original draft, Methodology; **Zhixin Li:** Writing – review & editing, Funding acquisition; **Shuni Jiang:** Visualization; **Canlong Zhang:** Validation; **Huifang Ma:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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